

Mitigating the Impact of Channel Aging in Cell-Free MIMO Systems Using a Channel Predictor Based on Extended Kalman Filter

A. Regilane L. Paiva¹, Walter C. Freitas Jr.², Member, IEEE, Roberto P. Antonioli³,
Yuri C. B. Silva⁴, Member, IEEE, and Gábor Fodor⁵, Fellow, IEEE

Abstract—Several previous works have pointed out that the channel aging effect may severely degrade the performance of wireless systems. This effect may be more pronounced in cell-free networks due to two main reasons. First, designing centralized precoders for cell-free (CF) systems requires the exchange of a large amount of channel state information (CSI) data between access points and a central processing unit, which may cause considerable processing delays. Secondly, the CF architecture requires keeping track of multiple aging channels, since user equipment devices are served by multiple cell sites. In this paper, we consider a CF multiple input multiple output (MIMO) system that uses joint coherence transmission and operates in the presence of channel propagation clustering under high Doppler frequencies. Unlike previous works, which focus on analyzing the impact of channel aging, this paper proposes a Kalman filter-based directional CSI prediction scheme that relies on a set of underlying P -th order autoregressive processes that correspond to the active links of the CF system. The extended Kalman filter is chosen with the purpose of predicting the channel path parameters. Using the predicted directional CSI, we design centralized hybrid precoding. Our results indicate that the proposed CSI prediction and precoding schemes outperform related benchmarks schemes and a 3rd order AR process model is a good engineering trade-off between model complexity and CSI prediction performance in practically relevant CF scenarios.

Index Terms—Autoregressive processes, cell-free, channel aging, Kalman filters, non-isotropic scattering.

I. INTRODUCTION

A. Background

THE main requirements of future generations of mobile communication networks include [1]: heterogeneity, pervasiveness, ultra-densification, spectral efficiency (SE), link reliability, energy efficiency, low latency, and integration of intelligent and positioning services. Such networks will bring together a diversity of communication technologies, air interfaces, infrastructure, architectures and applications, which will make them increasingly subject to channel propagation conditions and aging. The CF-massive multiple input multiple output (mMIMO) topology [2] emerges as a promising approach to increase the spectral and energy efficiency of mobile communication systems. The ultradensification, distribution and operation characteristics of CF-mMIMO systems facilitate the exploitation of favorable propagation and channel hardening properties in rich scattering environments [3], [4]. However, the theoretical analyses and research results are limited and do not strictly guarantee such properties under realistic operating and adverse channel conditions, such as sharp path loss differences [5], millimeter-wave (mmWave) operation and non-isotropic scattering [6], spatial correlation [7], imperfect channel state information (CSI) [8] and mainly channel aging [9], making their joint numeric analysis important. Channel aging is the gradual outdated (aging) of the CSI due to continuous channel variations caused by the relative movement between the receiving and transmitting antennas, as well as hardware impairments [10], [11], [12]. In contrast to the conventional block fading model, where channel samples are assumed to be uncorrelated between blocks and invariant within the coherence block, the channel aging fading model assumes temporal correlation and variation among channel samples along the transmission blocks. Doppler shift and CSI delays inherent in large CF networks, such as propagation and processing delay, accentuate the effects of channel aging [10]. In practice, there is a delay time between CSI acquisition and coherent data transmission or detection [13], even in open loop schemes. As the channel may change during this delay time, there will be a mismatch between CSI and actual channel, affecting the effectiveness of the precoders and detectors to canceling interference, causing degradation of SE that accumulates with time. Also, Doppler shift decreases the

Received 20 June 2024; revised 5 November 2024; accepted 6 January 2025. Date of publication 31 January 2025; date of current version 20 June 2025. The work of Gábor Fodor was supported by the EU Horizon Europe Program under Grant 101139176 6G-MUSICAL. This work was supported in part by Ericsson Research, Sweden, in part by Ericsson Innovation Center, Brazil, Technical Cooperation Contract UFC.50, in part by CNPq, in part by FUNCAP, in part by FUNCAP/Universal under Grant UNI-0210-00043.01.00/23, in part by CAPES - Finance Code 001, in part by CAPES/PRINT under Grant 88887.311965/2018-00, and in part by CNPq/INCT-Signals under Grant 406517/2022-3. The review of this article was coordinated by Prof. Kun Yang. (Corresponding author: A. Regilane L. Paiva.)

A. Regilane L. Paiva, Walter C. Freitas Jr., Roberto P. Antonioli, and Yuri C. B. Silva are with the Wireless Telecom Research Group, Federal University of Ceará, Fortaleza 60455-760, Brazil (e-mail: regilane@gtel.ufc.br; walter@gtel.ufc.br; antonioli@gtel.ufc.br; yuri@gtel.ufc.br).

Gábor Fodor is with the Ericsson Research, KTH Royal Institute of Technology, 16480 Stockholm, Sweden, and also with the Division of Decision and Control, KTH Royal Institute of Technology, 11428 Stockholm, Sweden (e-mail: gabor.fodor@ericsson.com).

Digital Object Identifier 10.1109/TVT.2025.3533969

channel coherence time by increasing small-scale variability, leading to fast spatio-temporal decorrelation among channel samples.

To alleviate congestion in the microwave spectrum, [14] and [15] have focused on analyzing and developing mmWave band solutions for CF-mMIMO. The setup composed of hybrid beamforming (HBF) architectures and centralized zero-forcing (ZF) precoder has been adopted therein aiming at high energy and spectral efficiency. The authors concluded that centralized ZF precoding outperforms maximum-ratio precoders in CF-mMIMO systems [13], [16]. However, centralized ZF-based data transmission requires more frequent exchanges of information between the access points (APs) and the central processing unit (CPU), resulting in increased CSI delay [17]. Furthermore, the high carrier frequencies of mmWave bands increase Doppler shift. In addition to conventional CF microwave setups, CF mmWave setups are even more susceptible to the degrading effects of channel aging.

B. Related Works

When local computing and changes in the physical air interface design, such as equalization, transmission time interval and waveform, are critically constrained and insufficient to reduce the delay and Doppler effect, or even when channel conditions are unfavorable, solutions based on Bayesian channel prediction can attenuate the impact of channel aging.

Some papers have addressed the channel aging effect in single- and multi-cell MIMO networks. In [11], a study of the channel aging effect in multi-cell multi-user (MU)-MIMO network operating in sub-6GHz bands was proposed and carried out. In addition, aiming to mitigate this effect, a Wiener filter (WF) is proposed to predict the future channel state from current and past pilots. Idealizing a Jake's isotropic-scattering model that determines the time correlation coefficient of the complex path gains, the spatially uncorrelated Rayleigh fading channel is modeled as a wide-sense stationary 1st order autoregressive (AR) process and then the filter coefficients are determined based on a P -order exponential correlation decay model. The main results indicate that increasing the MIMO system size remains beneficial for communication, even in the presence of channel aging, and the use of the predictor, whose effectiveness depends on the choice of order P , can partially overcome the effect of channel aging. In [18] it was demonstrated that channel aging does not decrease the power scaling law in single-cell co-located mMIMO, and the use of channel prediction based on linear WF does not improve the power scaling law. Kalman filter (KF) may be preferred over WF due to its ability to adapt to non-stationary systems and versatile modeling. In [19], a channel predictor based on P -order autoregressive moving average and linear KF was evaluated for a single-cell MU-MIMO system in sub-6GHz bands with the objective of reducing pilot spacing. Aiming at lower complexity of prediction models based on $AR(P)$ and linear KF, the coefficients of an $AR(2)$ model were optimized in [20] for a single-input single-output (SISO) spatially uncorrelated Rayleigh fading channel. Prediction of propagation parameters based on non-linear KF was studied in [21] as an alternative to the conventional method of predicting channel

coefficients based on linear KF, where the extended Kalman filter (EKF), unscented Kalman filter (UKF) and ensemble Kalman filter (EnKF) with $AR(1)$ were evaluated for a MIMO end-to-end communication system operating in mmWave. Unlike conventional pilot-aided prediction methods, in [22] a data-aided linear KF-based predictor is proposed for single-cell networks. In [23] the performance of a multilayer perceptron (MLP) predictor against a linear KF is evaluated on a single-cell network. To make MLP performance closer to that of KF, data pre-processing was adopted. It is challenging to deal with non-stationary processes using non-recurrent neural networks. Based on this, a long short-term memory (LSTM)-based predictors is proposed for SISO system in mmWave bands [24]. However, unlike methods based on signal processing, such as KF, these methods based on supervised learning may not guarantee robustness to noisy and insufficient measurements, interpretability and dynamic adaptation through parameter adjustments. Furthermore, machine learning methods require large volumes of data and life cycle management, as well as underfitting and overfitting control.

Recently, the channel aging effect has been investigated in CF-mMIMO systems on sub-6 GHz band [9], [17], [25]. The authors in [9], using multi-antenna APs, observed that the effect of channel aging on matched filter receiver decreases SE, but that CF networks may be more robust to this effect compared to conventional small-cell networks. It is also observed that channel spatial correlation can reduce the effect of channel aging and this in turn can reduce the effect of pilot contamination. In [25] the authors raise insights that the propensity for the effects of channel aging between CF and small-cell networks may depend on the values of Doppler shift and frame length, as well as precoding, so that the impact of channel aging can be higher in CF-mMIMO systems. In [17], the authors evaluate the impact of channel aging on centralized ZF precoding using initial perfect CSI and APs with single-antenna. Taking into account user equipment (UE) mobility, CSI delay and phase noise, it is observed that the impact on performance can be much more severe. Channel aging has been considered on mmWave for heterogeneous co-located MIMO in [26] and CF-mMIMO system in [27]. These works highlight the significant impact on network performance, such that mitigation solutions are needed, but they do not focus on combating this effect. In [28], an LSTM-based predictor is proposed for CF-MIMO in sub-6GHz bands [28], which is subject to practical challenges inherent to supervised learning.

The literature lacks solutions and analyses for mitigating channel aging in CF-MIMO. Furthermore, to the best of our knowledge, existing works do not address training and prediction of space-time CSI and generalization of directional or non-isotropic channel models that can influence the strict conclusions of the impact of channel aging on the SE in CF networks. General P -order AR process or $AR(P)$ models can be adjusted to generate spatially uncorrelated and correlated Rayleigh and Rician fading with an arbitrary space-time correlation function (STCF) [29], extending the application of stochastic filters, such as Kalman and Wiener, in channel estimation and prediction. Incorporating channel propagation clustering into channel aging mitigation studies allows for a more accurate and realistic analysis of the performance of these networks, providing

TABLE I
SUMMARY OF STATE OF THE ART RELATED TO CHANNEL PREDICTION AND CHANNEL AGING

Reference	Channel prediction	Support Non-stationary	Solution	Directional CSI	Channel Model	Network Analysis	Architecture
Truong and Heath [11]	WF-AR(P) pilot-aided tracking	no	near-optimal	no	- uncorrelated Rayleigh fading - Jakes time - AR(P) simulation [30] - fading from symbol to symbol	UL/DL multicell MU-MIMO	UCA digital sub-6 Ghz
Chui Li Kong et al. [18]	WF-AR(P) pilot-aided tracking	no	near-optimal	no	- uncorrelated Rayleigh fading - Jakes time - fading from symbol to symbol	DL single-cell and UL multi-cell mMIMO	ULA digital sub-6 Ghz
Kashyap et al. [19]	KF-ARMA(P, P) pilot-aided tracking	yes	near-optimal	no	- uncorrelated Rayleigh fading - Jakes time - fading from symbol to symbol	DL single-cell MU-MIMO	ULA digital sub-6 Ghz
El Hussein et al. [20]	KF-AR(2) pilot-aided tracking	yes	optimal	no	- uncorrelated Rayleigh fading	SISO	sub-6 Ghz
A. R. Paiva et al. [21]	EKF, UKF, EnKF-AR(1) pilot-aided tracking	yes	near-optimal	yes	- uncorrelated Rayleigh fading - AR(1) simulation	Point-to-point beamforming	UPA HYB mmWave
Kim et al. [23]	MLP pilot-aided predictor	no	black box	no	- uncorrelated Rayleigh fading - Jakes time - AR(P) simulation [30] - fading from frame to frame	UL single-cell MU-MIMO	UPA digital sub-6 Ghz
Chopra and Murthy [22]	KF-AR(1) data-aided tracking	yes	near-optimal	no	- uncorrelated Rayleigh fading - AR(1) simulation - fading from subframe to subframe	UL/DL single-cell MU-MIMO	ULA digital sub-6 Ghz
Wei J. and Hans D. [28]	LSTM pilot-aided	yes	black box	no	- uncorrelated Rayleigh fading	DL cell-free MIMO	single-antenna sub-6 Ghz
Varshney et al. [24]	LSTM pilot-aided	yes	black box	no	- uncorrelated Rayleigh fading - NYUSIM simulation [31]	SISO	mmWave
Present paper	EKF-AR(P) pilot-aided tracking	yes	near-optimal	yes	- uncorrelated and correlated Rayleigh and Rician fading based on GBSM - AR(P) simulation [30] - fading from symbol to symbol	DL cell-free MIMO	UPA HYB mmWave

valuable insights for the development of efficient solutions. The main related works with their characteristics and configurations are listed and compared with those of the present paper in Table I.

C. Novelty and Contributions

Motivated by the above observations, we propose an EKF-AR(P) based channel parameters predictor to mitigate the impact of channel aging in CF-MIMO systems for general fading models. The EKF model parameters are adjusted based on AR(P) and autocorrelation functions (ACFs). We explore the channel temporal correlation with AR(P) process model to predict the future channel state from current training pilots, and then we consider the clustered channel model. We adopted the estimation strategy discussed in existing works [21], [32], [33] in which sparsity, pilot based beamforming and beam codebook are considered for estimating the path parameters. This approach, instead of directly predicting the channel coefficients as done in the literature with linear KF [19], [20] and linear WF [11], uses EKF to predict the angles and the small-scale gain of sparse channel-independent paths, thus providing a directional CSI decomposed into spatial and gain information. Snapshots of angular and gain information of paths can be applied to the next generations of mobile networks and mmWave systems, such as: beam alignment and tracking decision-making [34], user scheduling schemes based on angular domain [35], positioning, sensing and including data collection for machine learning solutions in wireless networks [36], [37]. Also, such a strategy may have advantages in terms of backpropagation variables, information exchange and training overhead, since the number of parameters may be smaller than the number of antennas in propagation clustering, in addition to handling more detailed transition models.

The main contributions of this paper are:

- We extend the classical concept of channel tracking to predict path parameters in CF-mMIMO systems with general fading models by proposing an EKF-AR(P) based

predictor, enabling channel prediction in space and time. We numerically investigate the benefits of EKF-AR(P) based predictors in CF networks with centralized ZF-based hybrid precoder with realistic parameters, that vary continuously in space and time, in relation to the order P . For this purpose, we create a symbol-by-symbol simulation environment with random distribution of APs, UEs and scatterer clusters, geometry-based stochastic model (GBSM) channel, evolution of angles and small-scale gains based on AR, shadowing map and path loss from field measurements. We consider scenarios with channel aging, spatially uncorrelated and correlated Rayleigh and Rician fading, APs and UEs equipped with uniform planar array (UPA) as case studies. We observe that high P orders, such as $P > 5$, are unnecessary for the EKF, and that $P = 3$ meets the performance-complexity trade-off for the investigated scenarios.

- We compare the EKF predictor with the benchmarks: least squares (LS) estimator [38] and linear WF predictor of omnidirectional CSI [11], besides the ideal and aged CSI bounds. We extend the WF predictor in [11] to the most general case addressed in this paper, considering: AR of arbitrary order P , complex temporal coefficients, clustered paths and hybrid precoders.
- We evaluate the impact of space-time channel aging in relation to uplink training effective signal-to-noise ratio (SNR), CSI delay plus Doppler shift, spatial selectivity, Rician factor and block length. Our simulations show that the effect of channel aging is accentuated by high values of Doppler shift and CSI delay, which can make communication unsustainable without precautionary action. Notably, our results show that the EKF predictor can partially overcome the effects of channel aging, keeping communication more stable, improving performance in terms of downlink rate per user in conditions high frequency Doppler, even with insufficient measurements, as it provides an estimate of spatial and gain parameters, while the LS and WF methods provide estimates of the

channel coefficients less precisely, requiring greater computational effort. It is also shown that small random angular variations of the paths, which are often disregarded in the literature, can contribute to the increase of the channel aging effect. Furthermore, the directivity and strength of the line of sight (LoS) component can improve mitigation by prediction.

Considering the challenges posed by mmWave solutions, particularly regarding the cost and energy efficiency of distributed MIMO systems, we propose to adopt HBF architectures. Such architectures, with antenna arrays on both the AP and UE sides, provide beamforming gains that help overcome the severe path loss associated with short wavelengths and LoS blocking, as well as reduce co-channel interference at the training and transmission stage. The architectural cost is reduced by adopting a smaller number of mixed-signal devices compared to the number of antennas. Additionally, an analog codebook simplifies the phase shifter network by using predefined phase patterns, further reducing system complexity. This approach not only decreases the operational costs of the CF infrastructure but also reduces energy consumption.

D. Paper Outline and Notations

This paper is structured as follows. Section II describes the system model, where we define the models of the received signal, the parametric channel, the channel aging fading, the uplink training and downlink data transmission model. Section III describes the purpose of our predictor, as well as the prediction scheme. Section IV describes the simulation methodology and numerically evaluates the performance of the proposed model in relation to the benchmarks. Finally, Section V summarizes the main insights and conclusions.

Notations: $(\cdot)^\dagger$ and $(\cdot)^\top$ denote the Hermitian transpose and transpose of a matrix/vector, respectively. $|\cdot|$, $\|\cdot\|$ and $\|\cdot\|_F$ the absolute value of a scalar, the norm of a vector and the Frobenius norm of a matrix, respectively. $[\mathbf{A}]_{h,j}$ and $[\mathbf{a}]_h$ denote the (h,j) -th element of the matrix $\mathbf{A}$ and (h) -th element of the vector $\mathbf{a}$, respectively. x_y , $x^{(y)}$ and $x^{(y)2}$ denote subscript and superscript (in italic) of index y of x and the square of x superscript of index y , respectively. x_z^x denotes variable x of type y and z (not italic). $f[\cdot]$ denotes discrete argument function. $\mathbb{E}\{\cdot\}$, $\text{diag}(\cdot)$ and $\text{tr}(\cdot)$ denote the statistical expectation, diagonal and trace of the matrix, respectively. $\mathbf{0}$ and $\mathbf{1}$ denote a vector of 0's and 1's, respectively. We list the main variables and parameters defined throughout the article in Table II.

II. SYSTEM MODEL

A CF system composed of M APs and K UEs operating on carrier frequency f_c (Hz) is considered. Each AP, equipped with a UPA of $N_{\text{AN}}^{\text{AP}}$ antenna elements, is connected to a CPU via error-free fronthaul links with infinite capacity. The APs coherently serves on the same time-frequency resource all K UEs, each equipped with a UPA of $N_{\text{AN}}^{\text{UE}}$ antenna elements.

We consider a single frame divided into time-slots, each consisting of τ_s symbols. The temporal models are discrete and described in relation to the symbol index i and time-slot index n , such that n_i represents the i -th symbol of the n -th time-slot. The

TABLE II
MAIN VARIABLES AND PARAMETERS

Notation	Definition
M, K, m, k	the number of APs and UEs, AP and UE index, respectively.
$N_{\text{AN}}^{\text{AP}}, N_{\text{RF}}^{\text{AP}}, N_{\text{S}}, N_{\text{AN}}^{\text{UE}}, N_{\text{RF}}^{\text{UE}}$	the number of antenna elements, RF chains and streams from the APs, and the number of antenna elements and RF chains from the UEs, respectively.
L, l, P, p	the number of paths and path index, the order and index of the AR process coefficient, respectively.
$B_{\text{F}}, B_{\text{W}}, D_{\text{S}}, D_{\text{T}}$	the number of training beams from the APs and UEs, EKF state and observation variable dimension, respectively.
n_i, n_t, τ_s, τ_s	the i -th symbol of the n -th time-slot of the frame, n_t -th tap of the EKF, delay and transmission block length, respectively.
$g_{m,k}^{(l,n_i)}, \phi_{A,m,k}^{(l,n_i)}, \phi_{D,m,k}^{(l,n_i)}, \theta_{A,m,k}^{(l,n_i)}, \theta_{D,m,k}^{(l,n_i)}, \beta_{m,k}, \alpha_{m,k}$	complex channel gain, azimuth AoA and AoD, elevation AoA and AoD, large-scale and small-scale gain, respectively.
$\rho_{m,k}^{(l,\tau_s)}, f_{\text{ND},k}$	time correlation coefficient, normalized maximum Doppler frequency shift from the channel and EKF, respectively.

communication is divided into two phases: uplink training and downlink data transmission. The downlink channel is considered to be reciprocal to the uplink channel, an assumption that holds true in time division duplex (TDD) mode. The lengths of the time intervals in terms of symbols are: τ_d (downlink data block length), τ_q (uplink pilot block length), τ_s (transmission block length), such that $\tau_s = \tau_d + \tau_q$ and τ_δ (spacing delay), with $\tau(\cdot) \in \mathbb{N}$. The duration in units of seconds of a time-slot is defined based on symbol period T_s .

A. Signal Model

Due to the large number of APs deployed in CF environments, low-cost HBF architectures are needed to reduce the cost and energy consumption. Thus, consider that each AP contains $N_{\text{RF}}^{\text{AP}}$ radio frequency (RF) chains in a fully-connected HBF structure and transmits N_s streams, such that $N_s \leq N_{\text{RF}}^{\text{AP}} < N_{\text{AN}}^{\text{AP}}$. Also consider that each UE contains $N_{\text{RF}}^{\text{UE}}$ RF chains in a fully-connected HBF structure, such that $N_{\text{RF}}^{\text{UE}} < N_{\text{AN}}^{\text{UE}}$, receiving in a single-stream mode. Aiming at better performance in correlated channels, consider discrete Fourier transform (DFT)-based RF-codebooks $\mathcal{B}_{\text{UE}} = \{\mathbf{w}_{\text{RF}}\}_{|\mathcal{B}_{\text{UE}}|}$ and $\mathcal{B}_{\text{AP}} = \{\mathbf{f}_{\text{RF}}\}_{|\mathcal{B}_{\text{AP}}|}$, which correspond to the RF combiner and RF precoder, respectively. Assuming limited resolution analog phase shifters, such that the numbers B_{AP} and B_{UE} of beam patterns are equal to the numbers $N_{\text{AN}}^{\text{AP}}$ and $N_{\text{AN}}^{\text{UE}}$ of antenna elements, and constant-amplitude constraints, we adopt the primary DFT codebook for UPA $\mathcal{B}_{\text{AP(UE)}}$ as in [39]. For simplicity, APs and UEs have the same codebook, used for both channel training and data transmission.

Since each AP can transmit $N_s \leq N_{\text{RF}}^{\text{AP}}$ streams, each AP will serve a number of UEs $K = N_s$ (single stream per UE). Let $\mathbf{s}_d \in \mathbb{C}^{N_s \times 1}$ be the data symbol signal to be transmitted subject to total power constraints, where $\mathbb{E}\{\mathbf{s}_d \mathbf{s}_d^\dagger\} = \frac{1}{N_s} \mathbf{I}_{N_s}$. The m -th AP encodes the data symbols through the baseband precoder component $\mathbf{F}_{\text{BB},m} = [\mathbf{f}_{\text{BB},m,k}]_{k=1}^{N_s} \in \mathbb{C}^{N_{\text{RF}}^{\text{AP}} \times N_s}$, then beamforms the signal through the RF precoder component $\mathbf{F}_{\text{RF},m} = [\mathbf{f}_{\text{RF},m,h}]_{h=1}^{N_{\text{AN}}^{\text{AP}}} \in \mathbb{C}^{N_{\text{AN}}^{\text{AP}} \times N_{\text{RF}}^{\text{AP}}}$ for $\mathbf{f}_{\text{RF},m,h} \in \mathcal{B}_{\text{AP}}$.

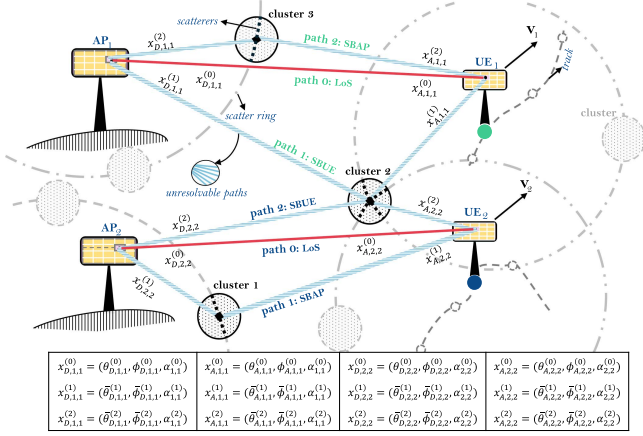


Fig. 1. Example of two 3-scatterer cluster channels in a CF network: one between AP-1 and UE-1, and another between AP-2 and UE-2. There are four rings of scatterers: two around the APs (one per AP) and two around the UEs (one per UE).

The data symbol signals $\mathbf{y}_k^{(n_i)} \in \mathbb{C}^{N_{AN}^{UE} \times 1}$ received by the k -th UE at time symbol n_i are given by

$$\mathbf{y}_k^{(n_i)} = \sum_{m=1}^M \sqrt{\rho_d} \mathbf{H}_{DL,m,k}^{(n_i)} \mathbf{F}_{RF,m}^{(n)} \mathbf{f}_{BB,m,k}^{(n)} s_{d,k}^{(n_i)} + \sum_{\substack{k'=1 \\ k' \neq k}}^K \sum_{m=1}^M \sqrt{\rho_d} \mathbf{H}_{DL,m,k'}^{(n_i)} \mathbf{F}_{RF,m}^{(n)} \mathbf{f}_{BB,m,k'}^{(n)} s_{d,k'}^{(n_i)} + \mathbf{u}_{DL}^{(n_i)}, \quad (1)$$

where the first term is the intended signal, the second is the interfering signal, $\mathbf{u}_{DL} \sim \mathcal{CN}(\mathbf{0}, \sigma_{DL}^2 \mathbf{I}_{N_{AN}^{UE}})$ denotes the complex Gaussian reception noise vector and $s_{d,k}^{(n_i)}$ is the data symbol transmitted to the k -th UE at time symbol n_i . $\mathbf{H}_{DL,m,k} \in \mathbb{C}^{N_{AN}^{UE} \times N_{AN}^{AP}}$ is the downlink channel matrix of the (m, k) -th AP-UE pair and the constraint $\|\mathbf{F}_{RF,m}^{(n)} \mathbf{f}_{BB,m,k}^{(n)}\|_F^2 = N_s$ is imposed, where ρ_d is the downlink effective power that takes into account the array gain. The receiver can apply the RF combining vector $\mathbf{w} \in \mathbb{C}^{N_{AN}^{UE} \times 1}$ to retrieve the data symbol, where $\mathbf{w}_k = \frac{1}{\sqrt{N_{AN}^{UE}}} \mathbf{1}_{N_{AN}^{UE}}$, so that the UEs are configured in single-stream.

B. Space-Time Varying Channel Model

Consider an L -scatterer channel model for the (m, k) -th AP-UE pair formed by $L_{m,k}$ fixed dominant spatially resolvable propagation paths [40]. We adopt the geometric model of MIMO channels based on two-ring scatterers as described in [41]. For analytical simplicity, we assume that all channels are flat fading and have the same number of paths, i.e., $L_{m,k} = L$, and that the index l represents the path, which can be of type LoS, non-line of sight (NLoS) single-bounced at the AP side (SBAP) (where the signal spreads only through clusters located around the AP) and NLoS single-bounced at the UE side (SBUE) (where the signal spreads only through clusters located around the UE), as illustrated in Fig. 1. We also assume that the APs and UEs oscillator clocks are synchronized and ideal. The fading model components of the uplink flat fading channel $\mathbf{H}_{UL,m,k}^{(n_i)} \in \mathbb{C}^{N_{AN}^{AP} \times N_{AN}^{UE}}$ at the n_i -th symbol for the (m, k) -th AP-UE pair are written in

terms of the physical channel parameters $\mathbf{x}_{m,k}^{(l,n_i)}$ as [40, p. 365]:

$$\mathbf{H}_{UL,m,k}^{(n_i)} \stackrel{\text{def}}{=} \sum_{l=1}^L \mathbf{H}_{UL} \left(\mathbf{x}_{m,k}^{(l,n_i)} \right), \quad (2)$$

such that

$$\mathbf{H}_{UL} \left(\mathbf{x}_{m,k}^{(l,n_i)} \right) = g_{m,k}^{(l,n_i)} \mathbf{a}_{AP} \left(\theta_{A,m,k}^{(l,n_i)}, \phi_{A,m,k}^{(l,n_i)} \right) \mathbf{a}_{UE}^\dagger \left(\theta_{D,m,k}^{(l,n_i)}, \phi_{D,m,k}^{(l,n_i)} \right), \quad (3)$$

where $g_{m,k}^{(l,n_i)}$, $\phi_{A,m,k}^{(l,n_i)}$, $\phi_{D,m,k}^{(l,n_i)}$, $\theta_{A,m,k}^{(l,n_i)}$ and $\theta_{D,m,k}^{(l,n_i)}$ are the complex channel gain, azimuth angle of arrival (AoA) and angle of departure (AoD), elevation AoA and AoD of the l -th path at the n_i -th symbol for the (m, k) -th AP-UE pair, respectively, and $\mathbf{a}_{AP(UE)}(\cdot)$ is the response of the UPA. For convenience, we omit the array gain. We define the l -th path parameters vector of the (m, k) -th AP-UE pair in the n_i -th symbol as:

$$\mathbf{x}_{m,k}^{(l,n_i)} \stackrel{\text{def}}{=} \left[\phi_{A,m,k}^{(l,n_i)}, \phi_{D,m,k}^{(l,n_i)}, \theta_{A,m,k}^{(l,n_i)}, \theta_{D,m,k}^{(l,n_i)}, g_{m,k}^{(l,n_i)} \right]^\top. \quad (4)$$

The channel parameters vary from symbol to symbol, but have large-scale properties that we consider invariant along the frame, namely: large-scale gain $\beta_{m,k}$ of the (m, k) -th AP-UE pair, UE velocity $\mathbf{v}_k$ and position $\mathbf{r}_k$ of the k -th UE, path power factor $\gamma_{m,k}^{(l)}$ of the l -th path from the (m, k) -th AP-UE pair and the L dominant paths. Then, we define the complex gain $g_{m,k}^{(l,n_i)}$ of the l -th path in the n_i -th symbol for the (m, k) -th AP-UE pair as:

$$g_{m,k}^{(l,n_i)} = \sqrt{\beta_{m,k} \gamma_{m,k}^{(l)}} \alpha_{m,k}^{(l,n_i)}, \quad (5)$$

where $\alpha_{m,k}^{(l,n_i)}$ is the small-scale gain of the l -th path from the (m, k) -th AP-UE pair in the n_i -th symbol, defined in Section II-C. The complex gain $g_{m,k}$ is such that $\mathbb{E}\{\|\mathbf{H}_{UL,m,k}\|_F^2\} = \beta_{m,k}$.

The spatial signature of the model is simplified with a normalized steering vector for a UPA with N_{AN} antenna elements expressed by:

$$\mathbf{a}(\theta, \phi) = \frac{1}{\sqrt{N_{AN}}} [a_1(\theta, \phi), a_2(\theta, \phi), \dots, a_{N_{AN}}(\theta, \phi)]^\top, \quad (6)$$

with

$$a_h(\theta, \phi) = \exp(j\mathbf{k}^\top(\theta, \phi)\mathbf{r}_h),$$

$$\mathbf{k}(\theta, \phi) = \frac{2\pi}{\lambda_c} [\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta]^\top,$$

where $\mathbf{k}$ is the wave vector, θ and ϕ are the elevation and azimuth components of the transmission/reception direction, respectively, $\mathbf{r}_h$ is the column vector of conventional coordinates of the h -th element antenna relative to array reference point, uniformly spaced at half wavelength, and λ_c is the wavelength. We assume that the geometry of the arrays is known and that they are oriented in the yz -plane. Finally, we use the TDD reciprocity property to determine the downlink channel, i.e., $\mathbf{H}_{DL,m,k}^{(n_i)} = \mathbf{H}_{UL,m,k}^{(n_i)\dagger}$.

C. Fading Model and Channel Aging

Taking into account the high Doppler frequency shifts, as well as a rich and dense environment, let us consider that the angular parameters and the small-scale complex coefficients of the channel model vary smoothly from symbol to symbol within a time-slot. Such assumptions can lead to detection and beamforming errors due to channel aging.

Works on CF MIMO that do not consider mobility adopt a block-fading model in which the channel instance in a coherence block is considered invariant. However, when mobility of the AP-UE antenna array pair and the dynamic context of the environment are high, channels vary continuously, leading to the channel aging effect where the channel samples are different but still maintain correlation with each other over the transmission block. This temporal correlation between samples generates smooth transitions that can be used to predict the channel and mitigate aging.

Consider that τ_δ is the CSI delay between the instant that channel is estimated and the instant that precoder is applied. Therefore, τ_δ is the sum of processing and propagation delays along the transmission block. If the channel parameters were estimated at an instant n_i , for the given lag τ_δ , the aged channel parameters are $\mathbf{x}^{(n_i+\tau_\delta)}$. In order to model the time-correlated fading, it is common in the literature [9], [11] to assume a stationary AR(1) discrete process, which is used for generating the small-scale fading parameters. However, the AR(1) model may not be sufficient to fit every variation of arbitrary second-order statistics regarding the temporal correlation. Jake's time ACF, for example, has been tuned with methods based on AR(P) [29] and inverse discrete Fourier transform (IDFT) [42]. The method of generating fading channels with AR(P) is introduced in Section III-A.

The stochastic model for the azimuth ϕ and elevation θ components of the AoAs (subscript A) and AoDs (subscript D) at n_i -th symbol for the l -th path from the (m, k) -th AP-UE pair are given by a Gaussian random walk [43]:

$$\begin{aligned}\theta_{A(D),m,k}^{(l,n_i+\tau_\delta)} &= \theta_{A(D),m,k}^{(l,n_i)} + u_{\theta_{A(D),m,k}}^{(l,n_i+\tau_\delta)}, \\ \phi_{A(D),m,k}^{(l,n_i+\tau_\delta)} &= \phi_{A(D),m,k}^{(l,n_i)} + u_{\phi_{A(D),m,k}}^{(l,n_i+\tau_\delta)},\end{aligned}\quad (7)$$

where $u_{\theta_{A(D),m,k}}^{(l,n_i+\tau_\delta)} \sim \mathcal{N}(0, \sigma_{\theta,m,k}^{(l)2})$ and $u_{\phi_{A(D),m,k}}^{(l,n_i+\tau_\delta)} \sim \mathcal{N}(0, \sigma_{\phi,m,k}^{(l)2})$. Since $\sigma_{\theta,m,k}^{(l)2}$ and $\sigma_{\phi,m,k}^{(l)2}$ control the temporal variation of angles, they may depend on: mobility, rotation, wobbling and shaking of UEs and scatterers, distance from AP to UE, among others [31], [44], [45], [46]. For simplicity, we consider that all paths have the same fixed angular variation, such that $\sigma_{\theta,m,k}^{(l)2} = \sigma_\theta^2$ and $\sigma_{\phi,m,k}^{(l)2} = \sigma_\phi^2$. Note that (7) is non-stationary, so that its second-order statistic grows indefinitely. On the other hand, the angular variations of the paths are considered very slow, so that within a transmission block we consider the autocorrelation to be time-invariant. (7) will be part of the EKF state transition matrix in Section III-B.

D. Uplink Training

The sequence of orthogonal pilot used by the k -th UE is $\Phi_k \in \mathbb{C}^{\tau_p \times N_{\text{RF}}^{\text{UE}}}$, such that $\tau_p \geq K N_{\text{RF}}^{\text{UE}}$ (no pilot reuse) and

$\Phi_k^\dagger \Phi_k = \mathbf{I}_{N_{\text{RF}}^{\text{UE}}}$. The fading parameters are assumed to be piecewise constant over a frequency coherence interval and the column $(\Phi_k)_{:,j}$ is arranged along the resource block bandwidth which is smaller than the coherence bandwidth.

The architectures of the AP and UE support $N_{\text{RF}}^{\text{AP}}$ and $N_{\text{RF}}^{\text{UE}}$ RF chains, respectively, so that it is possible to train up to $N_{\text{RF}}^{\text{AP}} N_{\text{RF}}^{\text{UE}}$ beams pairs simultaneously per symbol. If the training beamset $\mathcal{F} \subseteq \mathcal{B}_{\text{AP}}$ and $\mathcal{W} \subseteq \mathcal{B}_{\text{UE}}$, such that $|\mathcal{F}| = B_{\text{F}} \leq |\mathcal{B}_{\text{AP}}|$ and $|\mathcal{W}| = B_{\text{W}} \leq |\mathcal{B}_{\text{UE}}|$, are used by all APs and UEs for channel training respectively, the sets can be divided into $N_{\text{F}} = \lceil \frac{B_{\text{F}}}{N_{\text{RF}}^{\text{AP}}} \rceil$ and $N_{\text{W}} = \lceil \frac{B_{\text{W}}}{N_{\text{RF}}^{\text{UE}}} \rceil$ unique parts, like $\mathcal{F} = \{\mathbf{F}_{\text{RF}}^{(h)}\}_{h=1}^{N_{\text{F}}}$ and $\mathcal{W} = \{\mathbf{W}_{\text{RF}}^{(j)}\}_{j=1}^{N_{\text{W}}}$, then $\tau_q = N_{\text{F}} N_{\text{W}}$ pilot symbols are allocated.

Each AP receives by the h -th training sub-beamset $\mathbf{F}_{\text{RF}}^{(h)} \in \mathbb{C}^{N_{\text{AN}}^{\text{AP}} \times N_{\text{RF}}^{\text{AP}}}$ in the i -th pilot symbol of the n -th time-slot the mutual orthogonal pilot sequences $\{\Phi_k\}_{k=1}^K$ from j -th training sub-beamset $\mathbf{W}_{\text{RF}}^{(j)} \in \mathbb{C}^{N_{\text{AN}}^{\text{UE}} \times N_{\text{RF}}^{\text{UE}}}$. We can formulate the training as a function of the indices h and j as:

$$\mathbf{S}_{p,k}^{(j)} = \sqrt{\tau_p \varrho_p} \mathbf{W}_{\text{RF}}^{(j)} \Phi_k^\dagger \quad (8)$$

and

$$\mathbf{Y}_m^{(h,j)} = \mathbf{F}_{\text{RF}}^{(h)\dagger} \left(\sum_{k=1}^K \mathbf{H}_{\text{UL},m,k}^{(n_i)} \mathbf{S}_{p,k}^{(j)} + \mathbf{U}_{\text{UL},m}^{(n_i)} \right), \quad (9)$$

where ϱ_p is the uplink effective pilot power, $\mathbf{U}_{\text{UL}} \sim \mathcal{CN}(\mathbf{0}, \sigma_{\text{UL}}^2 \mathbf{I}_{N_{\text{AN}}^{\text{AP}} \tau_p})$ is the spatially white Gaussian noise, $\mathbf{S}_{p,k}^{(j)} \in \mathbb{C}^{N_{\text{AN}}^{\text{UE}} \times \tau_p}$ is the pilot matrix transmitted by the k -th UE and $\mathbf{Y}_m^{(h,j)} \in \mathbb{C}^{N_{\text{RF}}^{\text{AP}} \times \tau_p}$ is the pilot matrix received by the m -th AP. The training signal can then be retrieved by:

$$\begin{aligned}\tilde{\mathbf{Y}}_{m,k}^{(h,j)} &= \frac{1}{\sqrt{\tau_p \varrho_p}} \mathbf{Y}_m^{(h,j)} \Phi_k \\ &= \mathbf{F}_{\text{RF}}^{(h)\dagger} \mathbf{H}_{\text{UL},m,k}^{(n_i)} \mathbf{W}_{\text{RF}}^{(j)} \\ &\quad + \sum_{k' \neq k}^K \mathbf{F}_{\text{RF}}^{(h)\dagger} \mathbf{H}_{\text{UL},m,k'}^{(n_i)} \mathbf{W}_{\text{RF}}^{(j)} \Phi_{k'}^\dagger \Phi_k \\ &\quad + \frac{1}{\sqrt{\tau_p \varrho_p}} \mathbf{F}_{\text{RF}}^{(h)\dagger} \mathbf{U}_{\text{UL},m}^{(n_i)} \Phi_k \in \mathbb{C}^{N_{\text{RF}}^{\text{AP}} \times N_{\text{RF}}^{\text{UE}}}. \quad (10)\end{aligned}$$

The collection of training signals received with $D_{\bar{y}} = B_{\text{F}} B_{\text{W}}$ measurement samples of the (m, k) -th AP-UE channel in the n -th time-slot is defined as:

$$\bar{\mathbf{Y}}_{m,k} = \left[\tilde{\mathbf{Y}}_{m,k}^{(h,j)} \right]_{1 \leq h \leq N_{\text{F}}, 1 \leq j \leq N_{\text{W}}}, \quad (11)$$

where $\tilde{\mathbf{Y}}_{m,k} \in \mathbb{C}^{B_{\text{F}} \times B_{\text{W}}}$, which will be the observation variable in channel prediction based on EKF in Section III-C, which will generate $\hat{\mathbf{H}}_{\text{UL},m,k}^{(n_i+\tau_\delta)}$.

E. Downlink Data Transmission

We utilize low complexity HBF precoders with centralized ZF baseband processing based on [14]. We will predict the precoder for the next data block, which starts at the symbol $i = \tau_q + 1$ of the time-slot $n + 1$, as will be described in Section III-E. Thus, consider that $\tau_\delta = \tau_s$, that is, a delay time-slot. Based on (1),

the downlink SE of the k -th UE in time-slot $(n+1)$ is given by (12) shown at the bottom of the page.

III. PROPAGATION PARAMETERS PREDICTION BASED ON EKF-AR(P)

In this section, we develop a directional CSI predictor based on EKF and AR(P), capable of handling a variety of STCF channel models describing propagation parameters fading. Aiming at the correlation matching criterion, we tune the time ACF of the small-scale gains through an AR(P) to create a smooth state transition model that works as a blind state predictor. The AR(P) resembles an infinite impulse response filter, generating independent sample sequences of small-scale gains adhering to the desired ACFs. The variations of the angular parameters are adjusted with an AR(1) with a unitary coefficient. If channel variation is deterministic or quasi-deterministic, the model may be accurate. However, under realistic stochastic channel variations, assumed to be additive Gaussians in this paper, the AR model becomes a coarse state predictor requiring periodic error adjustment. For this tuning purpose, we propose a data assimilation process with EKF. The idea is to utilize the results of a spatial scanning process (beam search) of the channel assisted by pilots, capturing spatial information in real-time, to minimize AR(P) prediction error. This can be achieved through a nonlinear observation model and KF gains or adaptive coefficients.

The section then proceeds as follows: First, we present the AR fading generator, utilized both for creating the state transition model and designing our channel simulator for numerical results. Second, we construct our state transition matrix using the AR(P) coefficients. Third, we develop our observation model based on the parametric channel model and the pilots' a priori knowledge. Fourth, considering the a priori knowledge of the uncertainty statistics of the problem and the Jacobian matrix of the observation model, we calculate the near-optimal filter gains. Fifth, we employ our EKF-AR(P) based predictor to mitigate channel aging. We conclude the section with the EKF-AR(P) startup strategy.

A. AR-Based Small-Scale Fading

Assuming that the time ACF of the small-scale gain is known and which, for analytical simplification, is the same for each (m, k) -th channel, it is possible to determine the coefficients of a complex AR(P) process that closely approximate the generator of sequences of samples with time correlation. Considering the symbol as the time unit of our discrete channel, the small-scale gain $\alpha_{m,k}^{(l,n_i)} \stackrel{\text{def}}{=} \alpha_{m,k}^{(l)}[n_i]$ for each independent path l of the (m, k) -th AP-UE pair in the symbol n_i can be estimated by:

$$\alpha_{m,k}^{(l,n_i)} = \sum_{p=1}^P a_p^{(l)} \alpha_{m,k}^{(l,n_i-p)} + u_{\alpha,m,k}^{(n_i)}, \quad (13)$$

where P is the filter order, $a_p^{(l)} \in \mathbb{C}$ are the filter coefficients for the l -th path and $u_{\alpha,m,k} \sim \mathcal{CN}(0, \sigma_{\alpha,m,k}^{(l)2})$ is a circular complex white Gaussian noise (uncorrelated real and imaginary components) for the (m, k) -th AP-UE pair. The coefficients $a_p^{(l)}$ are determined via the Yule-Walker equations by the method in [29], where a bias ϵ is added to the small-scale gain variance to avoid ill conditioning correction matrix.

To implement the EKF state transition model, we computed the P coefficients $\mathbf{a} = [a_1, a_2, \dots, a_P]^\top$ such that:

$$\mathbf{a}^{(l)} = (\hat{\mathbf{Q}}_{\alpha\alpha}^{(l)})^{-1} \mathbf{q}^{(l)}, \quad (14)$$

where $\hat{\mathbf{Q}}_{\alpha\alpha} \in \mathbb{R}^{P \times P}$ is the autocorrelation matrix of the small-scale complex gain $\alpha^{(l)}$ given by:

$$\hat{\mathbf{Q}}_{\alpha\alpha}^{(l)} = [Q_{hj}]_{1 \leq h, j \leq P} \text{ and } Q_{hj} = \begin{cases} \hat{Q}_{\alpha\alpha}^{(l, |h-j|)} + \epsilon, & \text{if } h = j \\ \hat{Q}_{\alpha\alpha}^{(l, |h-j|)}, & \text{if } h > j, \\ \hat{Q}_{\alpha\alpha}^{(l, |h-j|)*}, & \text{if } h < j \end{cases}$$

such that $\hat{Q}_{\alpha\alpha}^{(l, \tau)}$ is the estimated time ACF, ϵ is a very small bias used to avoid singularity in the matrix inversion $\hat{\mathbf{Q}}_{\alpha\alpha}^{(l)}$ [29] and the variance vector $\mathbf{q}^{(l)} = [\hat{Q}_{\alpha\alpha}^{(l,1)}, \dots, \hat{Q}_{\alpha\alpha}^{(l,P)}]^\top$. The driving noise variance σ_α^2 is given by:

$$\sigma_\alpha^{(l)2} = \hat{Q}_{\alpha\alpha}^{(l,0)} - \sum_{p=1}^P a_p^{(l)} \hat{Q}_{\alpha\alpha}^{(l,p)*}. \quad (15)$$

B. EKF State Model

To design the multidimensional complex filter, we adopt the vector EKF structure, where all variables are concatenated within a single state vector $\bar{\mathbf{x}}$. Consider the following variables in the tap n_t :

$$\boldsymbol{\theta}_D^{(n_t)} = [\hat{\theta}_D^{(1,n_t)}, \dots, \hat{\theta}_D^{(L,n_t)}], \boldsymbol{\theta}_A^{(n_t)} = [\hat{\theta}_A^{(1,n_t)}, \dots, \hat{\theta}_A^{(L,n_t)}],$$

$$\boldsymbol{\phi}_D^{(n_t)} = [\hat{\phi}_D^{(1,n_t)}, \dots, \hat{\phi}_D^{(L,n_t)}], \boldsymbol{\phi}_A^{(n_t)} = [\hat{\phi}_A^{(1,n_t)}, \dots, \hat{\phi}_A^{(L,n_t)}],$$

$$\mathbf{g}_{\text{Re}}^{(n_t)} = \Re \left([\hat{g}^{(1,n_t)}, \dots, \hat{g}^{(L,n_t)}] \right)$$

and

$$\mathbf{g}_{\text{Im}}^{(n_t)} = \Im \left([\hat{g}^{(1,n_t)}, \dots, \hat{g}^{(L,n_t)}] \right),$$

$$R_k^{(n+1)} = \frac{1}{\tau_s} \sum_{i=\tau_q+1}^{\tau_s} \log_2 \left(1 + \frac{\left| \sum_{m=1}^M \sqrt{\frac{g_d}{N_s}} \mathbf{w}_{m,k}^{(n+1)\dagger} \mathbf{H}_{\text{DL},m,k}^{((n+1)_i)} \mathbf{F}_{\text{RF},m}^{(n+1)} \mathbf{f}_{\text{BB},m,k}^{(n+1)} \right|^2}{\sum_{k'=1, k' \neq k}^K \left| \sum_{m=1}^M \sqrt{\frac{g_d}{N_s}} \mathbf{w}_{m,k'}^{(n+1)\dagger} \mathbf{H}_{\text{DL},m,k'}^{((n+1)_i)} \mathbf{F}_{\text{RF},m}^{(n+1)} \mathbf{f}_{\text{BB},m,k'}^{(n+1)} \right|^2 + \sigma_{\text{DL}}^2} \right)}. \quad (12)$$

where $\Re(\cdot)$ and $\Im(\cdot)$ the real and imaginary part functions, $\hat{\theta}$ denotes estimated elevation components, $\hat{\phi}$ are the estimated azimuth components, $\hat{g}$ is the estimated path gain and L is the number of dominant paths. We define the state vector $\bar{\mathbf{x}}^{(n_t)}$ of order P as:

$$\bar{\mathbf{x}}^{(n_t)} = [\phi_A^{(n_t)}, \phi_D^{(n_t)}, \theta_A^{(n_t)}, \theta_D^{(n_t)}, \bar{\mathbf{g}}^{(n_t)}]^\top \in \mathbb{R}^{D_{\bar{\mathbf{x}}} \times 1}, \quad (16)$$

where

$$\bar{\mathbf{g}}^{(n_t)} = [\mathbf{g}^{(n_t)}, \mathbf{g}^{(n_t-1)}, \dots, \mathbf{g}^{(n_t-P+1)}]^\top \in \mathbb{R}^{2LP \times 1}$$

and

$$\mathbf{g}^{(n_t)} = [\mathbf{g}_{\text{Re}}^{(n_t)}, \mathbf{g}_{\text{Im}}^{(n_t)}]^\top \in \mathbb{R}^{2L \times 1},$$

such that $\bar{\mathbf{x}}^{(n_t)}$ has dimension $D_{\bar{\mathbf{x}}} = 4L + 2LP$.

Given the channel time evolution models defined by (7) and (13), we structure the EKF-AR(P) state transition equation as:

$$\bar{\mathbf{x}}^{(n_t+1)} = \bar{\mathbf{A}}\bar{\mathbf{x}}^{(n_t)} + \mathbf{u}_{\bar{\mathbf{x}}}^{(n_t+1)}, \quad (17)$$

where $\bar{\mathbf{x}}^{(n_t+1)}$ is a posteriori prediction and $\bar{\mathbf{A}}$ is the state transition matrix given by:

$$\bar{\mathbf{A}} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{0}_{4L \times 2LP} \\ \mathbf{0}_{2LP \times 4L} & \mathbf{A}_2 \end{bmatrix} \in \mathbb{R}^{D_{\bar{\mathbf{x}}} \times D_{\bar{\mathbf{x}}}}, \quad (18)$$

such that

$$\mathbf{A}_1 = \mathbf{I}_{4L} \in \mathbb{R}^{4L \times 4L},$$

$$\mathbf{A}_2 = \begin{bmatrix} \bar{\mathbf{A}}_{2,1} & \bar{\mathbf{A}}_{2,2} & \cdots & \bar{\mathbf{A}}_{2,P-1} & \bar{\mathbf{A}}_{2,P} \\ \mathbf{I}_{2L} & \mathbf{0}_{2L} & \cdots & \mathbf{0}_{2L} & \mathbf{0}_{2L} \\ \mathbf{0}_{2L} & \mathbf{I}_{2L} & \cdots & \mathbf{0}_{2L} & \mathbf{0}_{2L} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0}_{2L} & \mathbf{0}_{2L} & \cdots & \mathbf{I}_{2L} & \mathbf{0}_{2L} \end{bmatrix} \in \mathbb{R}^{2LP \times 2LP},$$

and

$$\bar{\mathbf{A}}_{2,p} = \begin{bmatrix} \Re(\mathbf{A}_{2,p}) & -\Im(\mathbf{A}_{2,p}) \\ \Im(\mathbf{A}_{2,p}) & \Re(\mathbf{A}_{2,p}) \end{bmatrix} \in \mathbb{R}^{2L \times 2L},$$

where $\mathbf{A}_{2,p} \in \mathbb{C}^{L \times L}$ is the coefficient matrix of the channel generator AR filter. For uncorrelated channels, it follows that $\mathbf{A}_{2,p} = \text{diag}([a_p^{(1)}, \dots, a_p^{(L)}])$, where $a_p^{(l)}$ comes from (14). The matrix $\mathbf{A}_2$ is a linear transformation with a lower shift matrix effect. It roughly predicts the future state by superimposing (submatrices $\mathbf{A}_{2,p}$ from the first line) and shifting (block matrix from line 2 to end) the current and past states. The vector $\mathbf{u}_{\bar{\mathbf{x}}}$ is the innovation process given by:

$$\mathbf{u}_{\bar{\mathbf{x}}}^{(n_t)} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_{\mathbf{u}_{\bar{\mathbf{x}}}}) \in \mathbb{R}^{D_{\bar{\mathbf{x}}} \times 1}, \quad (19)$$

such that

$$\mathbf{Q}_{\mathbf{u}_{\bar{\mathbf{x}}}} = \bar{\mathbf{B}}\mathbf{Q}_{\mathbf{u}_{\mathbf{x}}} \bar{\mathbf{B}}^\top \in \mathbb{R}^{D_{\bar{\mathbf{x}}} \times D_{\bar{\mathbf{x}}}},$$

where $\mathbf{Q}_{\mathbf{u}_{\mathbf{x}}}$ is the covariance matrix of the process noise to the state vector

$$\mathbf{x}^{(n_t)} = [\phi_A^{(n_t)}, \phi_D^{(n_t)}, \theta_A^{(n_t)}, \theta_D^{(n_t)}, \mathbf{g}^{(n_t)}]^\top \in \mathbb{R}^{6L \times 1}$$

and

$$\bar{\mathbf{B}} = [\mathbf{I}_{6L}, \mathbf{0}_{6L}, \dots, \mathbf{0}_{6L}]^\top \in \mathbb{R}^{D_{\bar{\mathbf{x}}} \times 6L}.$$

Note that the dimension of the state vector does not depend on the number of antennas, but on the number of path parameters. In terms of backpropagation variables and exchange of information, there will be a dimensional reduction to MIMO propagation clustering when the number of parameters is less than the number of antennas.

C. EKF Observation Model

The non-linear observation model based on beam search training in n_t -th tap (10) is given by:

$$\begin{aligned} \hat{\mathbf{Y}}^{(n_t)} &\stackrel{\text{def}}{=} \mathbf{Y}(\hat{\mathbf{x}}^{(n_t)}) \\ &= \mathbf{F}_{\text{RF}}^\dagger \hat{\mathbf{H}}_{\text{UL}}^{(n_t)} \mathbf{W}_{\text{RF}} \in \mathbb{C}^{B_{\text{F}} \times B_{\text{W}}}, \end{aligned} \quad (20)$$

where $\hat{\mathbf{x}}^{(n_t)} \in \mathbb{C}^{6L \times 1}$ comes from the backpropagated state variable of the EKF, $\mathbf{F}_{\text{RF}} \in \mathbb{C}^{N_{\text{AN}}^{\text{AP}} \times B_{\text{F}}}$ and $\mathbf{W}_{\text{RF}} \in \mathbb{C}^{N_{\text{AN}}^{\text{UE}} \times B_{\text{W}}}$. However, it is necessary to linearize the observation model (20) for EKF implementation. A linear approximation is made by calculating the Jacobian matrix $\mathbf{C}^{(n_t)} \stackrel{\text{def}}{=} \mathbf{C}(\hat{\mathbf{x}}^{(n_t)}) \in \mathbb{C}^{D_{\bar{\mathbf{y}}} \times 6L}$ of $\hat{\mathbf{Y}}^{(n_t)}$ so that:

$$\mathbf{C}^{(n_t)} = [c_{h,j}(\hat{\mathbf{x}}^{(n_t)})]_{1 \leq h \leq D_{\bar{\mathbf{y}}}, 1 \leq j \leq 6L}, \quad (21)$$

with

$$c_{h,j}(\hat{\mathbf{x}}^{(n_t)}) = \frac{\partial \hat{y}_h(\hat{\mathbf{x}}^{(n_t)})}{\partial x_j}, \quad (22)$$

where $\hat{y}_h(\hat{\mathbf{x}}^{(n_t)})$ is an element of the matrix function $\hat{\mathbf{Y}}^{(n_t)}$ and h the linear index.

In order to deal with complex state values, we vectorized the

matrix $\hat{\mathbf{Y}}^{(n_t)}$ in form $\hat{\mathbf{y}}^{(n_t)} = \begin{bmatrix} \Re(\text{vec}(\hat{\mathbf{Y}}^{(n_t)})) \\ \Im(\text{vec}(\hat{\mathbf{Y}}^{(n_t)})) \end{bmatrix} \in \mathbb{R}^{2D_{\bar{\mathbf{y}}} \times 1}$

and the training matrix (10) (observed variable) in $\tilde{\mathbf{y}}^{(n_t)} = \begin{bmatrix} \Re(\text{vec}(\bar{\mathbf{Y}}^{(n_t)})) \\ \Im(\text{vec}(\bar{\mathbf{Y}}^{(n_t)})) \end{bmatrix} \in \mathbb{R}^{2D_{\bar{\mathbf{y}}} \times 1}$. Finally, since the Jacobian matrix is computed only over the current state at tap n_t , we restructure $\mathbf{C}^{(n_t)}$ as:

$$\bar{\mathbf{C}}^{(n_t)} = \begin{bmatrix} \Re(\mathbf{C}^{(n_t)}) \\ \Im(\mathbf{C}^{(n_t)}) \end{bmatrix}, \mathbf{0}_{2D_{\bar{\mathbf{y}}} \times 2L(P-1)} \in \mathbb{R}^{2D_{\bar{\mathbf{y}}} \times D_{\bar{\mathbf{x}}}}. \quad (23)$$

D. EKF Gain

From the EKF primordial theory [47], our filter gain $\bar{\mathbf{K}} \in \mathbb{R}^{D_{\bar{\mathbf{x}}} \times 2D_{\bar{\mathbf{y}}}}$ is given by:

$$\bar{\mathbf{K}}^{(n_t)} = \bar{\mathbf{P}}^{(n_t)} \bar{\mathbf{C}}^{(n_t)\top} (\bar{\mathbf{C}}^{(n_t)} \bar{\mathbf{P}}^{(n_t)} \bar{\mathbf{C}}^{(n_t)\top} + \mathbf{Q}_{\mathbf{u}_{\bar{\mathbf{y}}}})^{-1}, \quad (24)$$

where $\bar{\mathbf{P}} \in \mathbb{R}^{D_{\bar{\mathbf{x}}} \times D_{\bar{\mathbf{x}}}}$ is the covariance matrix of the a posteriori prediction error of the state variable $\bar{\mathbf{x}}$ and $\mathbf{Q}_{\mathbf{u}_{\bar{\mathbf{y}}}} \in \mathbb{R}^{2D_{\bar{\mathbf{y}}} \times 2D_{\bar{\mathbf{y}}}}$ is

Algorithm 1: EKF Channel Prediction.

```

1: INITIAL VALUES:  $\hat{\mathbf{x}}_{0|0} \leftarrow \bar{\mathbf{x}}_0, \bar{\mathbf{P}}_{0|0} \leftarrow \bar{\mathbf{P}}_0$ 
2: INPUT:  $\hat{\mathbf{x}}_{n_t|n_t-1}, \bar{\mathbf{P}}_{n_t|n_t-1}, \bar{\mathbf{Y}}_n$ 
3: GAIN:
 $\hat{\mathbf{x}}_{n_t|n_t-1:n_t-P} \leftarrow [\hat{\mathbf{x}}_{n_t|n_t-1}]_{1:6L}$ 
 $\mathbf{C}_{n_t} \xleftarrow{\Re|\Im} \mathbf{C}(\hat{\mathbf{x}}_{n_t|n_t-1:n_t-P})$ 
 $\bar{\mathbf{C}}_{n_t} \leftarrow [\mathbf{C}_{n_t}, \mathbf{0}_{2D_{\bar{\mathbf{y}}} \times 2L(P-1)}]$ 
 $\bar{\mathbf{K}}_{n_t} \leftarrow \bar{\mathbf{P}}_{n_t|n_t-1} \bar{\mathbf{C}}_{n_t}^\top (\bar{\mathbf{C}}_{n_t} \bar{\mathbf{P}}_{n_t|n_t-1} \bar{\mathbf{C}}_{n_t}^\top + \mathbf{Q}_{u_{\bar{\mathbf{y}}} u_{\bar{\mathbf{y}}}})^{-1}$ 
4: ESTIMATOR:
 $\hat{\mathbf{y}}_{n_t} \xleftarrow{\Re|\Im} \mathbf{Y}(\hat{\mathbf{x}}_{n_t|n_t-1:n_t-P}) \quad (20)$ 
 $\tilde{\mathbf{y}}_{n_t} \xleftarrow{\Re|\Im} \bar{\mathbf{Y}}_n \text{ external variable}$ 
 $\mathbf{z}_{n_t} \leftarrow \tilde{\mathbf{y}}_{n_t} - \hat{\mathbf{y}}_{n_t} \text{ innovation process}$ 
 $\hat{\mathbf{x}}_{n_t|n_t} \leftarrow \hat{\mathbf{x}}_{n_t|n_t-1} + \bar{\mathbf{K}}_{n_t} \mathbf{z}_{n_t} \text{ state estimation}$ 
5: PREDICTION:  $\hat{\mathbf{x}}_{n_t+1|n_t} \leftarrow \bar{\mathbf{A}} \hat{\mathbf{x}}_{n_t|n_t}$ 
6: MSE ANALYSIS:
 $\bar{\mathbf{P}}_{n_t|n_t} \leftarrow (\mathbf{I}_{D_{\bar{\mathbf{x}}}} - \bar{\mathbf{K}}_{n_t} \bar{\mathbf{C}}_{n_t}) \bar{\mathbf{P}}_{n_t|n_t-1}$ 
 $\bar{\mathbf{P}}_{n_t+1|n_t} \leftarrow \bar{\mathbf{A}} \bar{\mathbf{P}}_{n_t|n_t} \bar{\mathbf{A}}^\top + \mathbf{Q}_{u_{\bar{\mathbf{x}}} u_{\bar{\mathbf{x}}}}$ 

```

the covariance matrix of the measurement noise. The computational complexity of the EKF comes mainly from calculating the gain, which deals with inverting a matrix of size $2D_{\bar{\mathbf{y}}} \times 2D_{\bar{\mathbf{y}}}$ and multiplying matrices of size $D_{\bar{\mathbf{x}}} \times D_{\bar{\mathbf{x}}}$ and $2D_{\bar{\mathbf{y}}} \times 2D_{\bar{\mathbf{y}}}$. Therefore, the number of beamforming pilots $B_F B_W$, the order P and the number of paths L influence the computational cost. The EKF pseudocode is described in Algorithm 1, where superscript n_t was transferred to subscript, the vectorization operation was represented by $\xleftarrow{\text{vec}}$ and the stacking of complex components by $\xleftarrow{\Re|\Im}$.

E. Channel Prediction

We assume that the CSIs ages continuously and that there is a gap (CSI delay), inherent to the network, between the training block and the data transmission block. This gap is related to the time required for all transmitters in the network to update their precoded data and start synchronized data transmission. The CSI delay is the budget of the sequential processes: transmission of the pilot sequence from the UEs to the APs, channel estimation and determination of precoder RF components in the APs, delivery of the CSIs from the APs to the CPU, processing of the ZF precoders centralized in the CPU, delivery of the precoded data from the CPU to the APs and AP synchronization for coherent data transmission, as illustrated in Fig. 2(a). Therefore, we can list two conventional possibilities: either the transmission stops-and-wait for new precoded data or it continues with the aged precoder. In both cases, there will be a drop in the transmission rate. To mitigate this drop, we predict the ZF precoders for the subsequent transmission block based on training on the current transmission block, such that data transmission starts with precoded data with minimized aging, as shown in Fig. 2(b). Thus, the prediction time distance τ_δ will be the spacing between data blocks.

The channel prediction method will be applied to each channel individually, thus we focus in the (m, k) -th AP-UE pair.

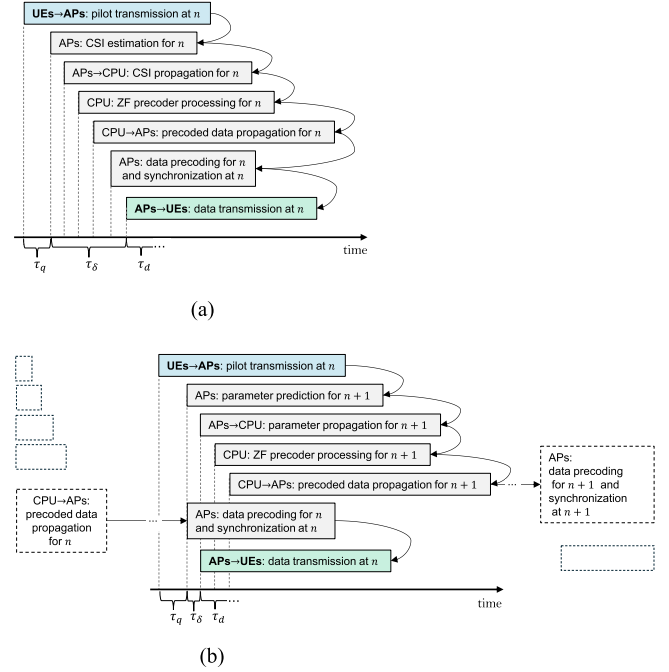


Fig. 2. Delays in a centralized ZF CF network and comparison between the conventional and proposed precoding method. (a) Conventional stop-and-wait. (b) Proposed.

Algorithm 1 was described for online single tap prediction and the prediction scheme is shown in Fig. 3. The spacing delay τ_δ is introduced in the normalized maximum Doppler shift so that: if $f_{ND,k} = \frac{f_c \|\mathbf{v}_k\|}{v_c} T_s$ (Hz) is the normalized maximum Doppler shift of the channel, where $\|\mathbf{v}_k\|$ is the UE absolute velocity (m/s), T_s is the symbol period (s) and v_c (m/s) is the speed of light, the normalized maximum Doppler shift of the EKF is $f_{ND}' = \tau_\delta f_{ND}$. This way, if tap n_t corresponds to the last training symbol $n_i = n_{\tau_q}$, $n_t + 1$ will correspond to the first data symbol $n_i + \tau_\delta$ that will be τ_δ symbols ahead. We assume that there is an $\text{EKF}_{m,k}(\cdot)$ for each AP-UE pair defined by the following parameters: $\hat{\mathbf{x}}_{m,k}^{(0)}$, $\bar{\mathbf{P}}_{m,k}^{(0)}$, $\mathbf{Q}_{u_{\bar{\mathbf{x}}} u_{\bar{\mathbf{x}}}, m, k}$ and $\mathbf{Q}_{u_{\bar{\mathbf{y}}} u_{\bar{\mathbf{y}}}, m}$. Therefore, the predicted channel used in determining the updated precoder will be $\hat{\mathbf{H}}_{UL, m, k}^{(n_i + \tau_\delta)} = \sum_{l=1}^L \mathbf{H}_{UL}(\hat{\mathbf{x}}_{m,k}^{(l, n_i + \tau_\delta)})$ where $\hat{\mathbf{x}}_{m,k}^{(n_i + \tau_\delta)} = [\hat{\mathbf{x}}_{n_t+1|n_t}]_{1:6L}$ and $\hat{\mathbf{x}}_{n_t+1|n_t} = \text{EKF}_{m,k}(\hat{\mathbf{x}}_{n_t|n_t-1}, \bar{\mathbf{P}}_{n_t|n_t-1}, \bar{\mathbf{Y}}_{m,k}^{(n_i)})$.

F. EKF-AR(P) Initialization

We assume that the initial angles are known, as well as the number of dominant paths,² large-scale gains and time ACF.

¹The first 6L elements of vector $\hat{\mathbf{x}}$.

²Classical methods such as orthogonal matching pursuit (OMP) [38] and Multiple Signal Classification (MUSIC) [48] can be used to determine the initial number of spatially resolvable paths, as well as their AoA and AoD. Methods like OMP, for example, can identify the number of spatially resolvable paths by estimating the L-sparse matrix of the channel through compressed sensing and channel sounding. These methods require high computational demand (large dictionaries of response arrangements) and long training sequences (refined training beams) for accurate estimates, in addition to being subject to ambiguity errors. Therefore, the prolonged use of these methods for updating the 3D estimates of the channel parameters at later times may exacerbate the effects of

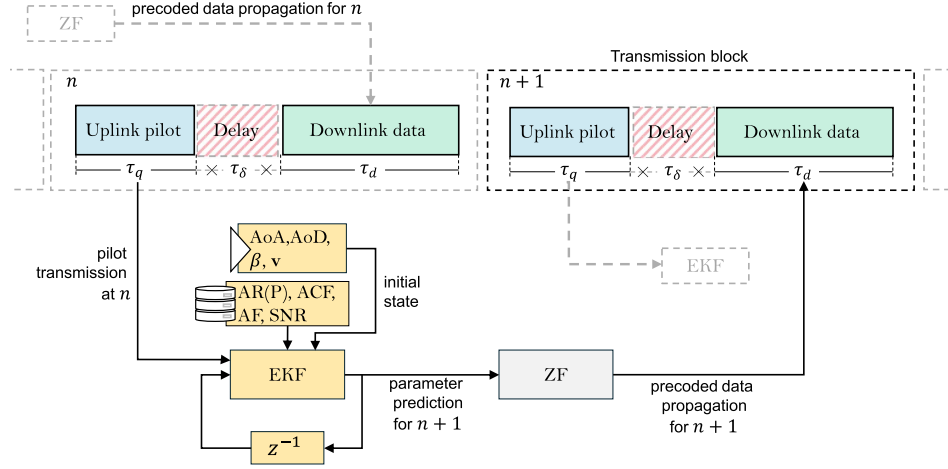
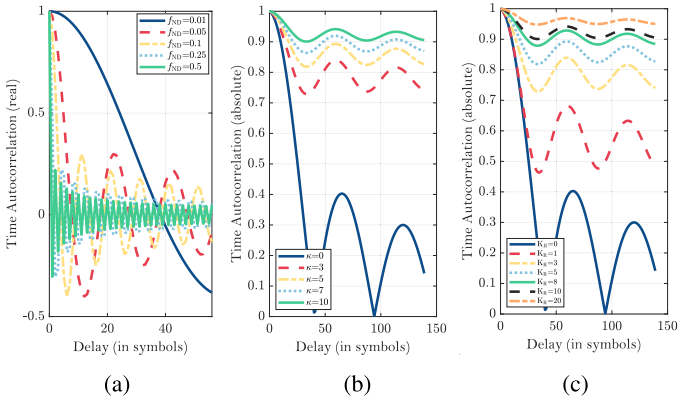
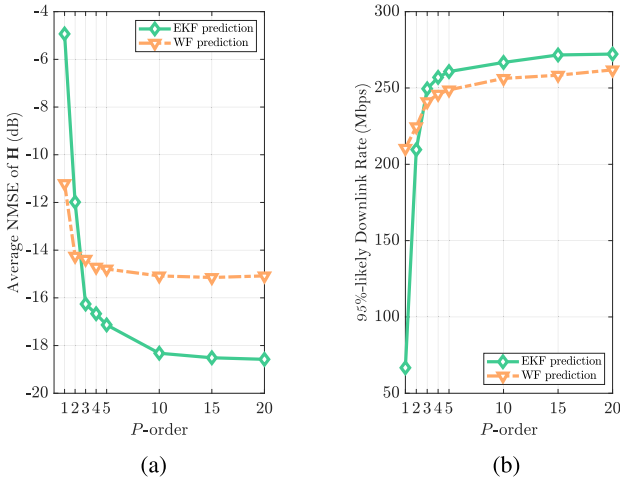


Fig. 3. Transmission block with channel aging mitigation.

Fig. 4. Influence of normalized Doppler frequency shift, angular dispersion and LoS component on channel coherence time. (a) f_{ND} . (b) κ . (c) K_R .Fig. 5. Influence of P order on EKF and WF performance for isotropic fading model. (a) Channel matrix error. (b) Downlink rate.

channel aging due to increased processing and propagation delays. Our method, which does not require dictionaries or very narrow beams, offers an adaptive and predictive solution based on channel autocorrelation functions, capable of

Considering uncorrelated paths, we define the initial state for each AP-UE pair (m, k) as:

$$\bar{\mathbf{x}}_{m,k}^{(0)} = [\hat{\mathbf{x}}_{m,k}^{(0)\top}, \mathbf{0}_{1 \times 2L(P-1)}]^\top \in \mathbb{R}^{D_{\bar{\mathbf{x}}} \times 1}, \quad (25)$$

where

$$\hat{\mathbf{x}}_{m,k}^{(0)} = [\hat{\phi}_{A,m,k}^{(0)}, \hat{\phi}_{D,m,k}^{(0)}, \hat{\theta}_{A,m,k}^{(0)}, \hat{\theta}_{D,m,k}^{(0)}, \mathbf{0}_{1 \times 2L}]^\top \in \mathbb{R}^{6L \times 1}.$$

The initial a posteriori prediction error covariance matrix is set up as:

$$\bar{\mathbf{P}}_{m,k}^{(0)} = \begin{bmatrix} \mathbf{0}_{4L} & \mathbf{0}_{4L \times 2LP} \\ \mathbf{0}_{2LP \times 4L} & \mathbf{P}_{2,m,k}^{(0)} \end{bmatrix} \in \mathbb{R}^{D_{\bar{\mathbf{x}}} \times D_{\bar{\mathbf{x}}}}, \quad (26)$$

where

$$\mathbf{P}_{2,m,k}^{(0)} = \begin{bmatrix} \mathbf{Q}_{\hat{\alpha}\hat{\alpha},m,k}^{(0)} & \mathbf{Q}_{\hat{\alpha}\hat{\alpha},m,k}^{(\tau_\delta)} & \cdots & \mathbf{Q}_{\hat{\alpha}\hat{\alpha},m,k}^{(\tau_\delta P - \tau_\delta)} \\ \mathbf{Q}_{\hat{\alpha}\hat{\alpha},m,k}^{(\tau_\delta)\dagger} & \mathbf{Q}_{\hat{\alpha}\hat{\alpha},m,k}^{(0)} & \cdots & \mathbf{Q}_{\hat{\alpha}\hat{\alpha},m,k}^{(\tau_\delta P - 2\tau_\delta)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{Q}_{\hat{\alpha}\hat{\alpha},m,k}^{(\tau_\delta P - \tau_\delta)\dagger} & \mathbf{Q}_{\hat{\alpha}\hat{\alpha},m,k}^{(\tau_\delta P - 2\tau_\delta)\dagger} & \cdots & \mathbf{Q}_{\hat{\alpha}\hat{\alpha},m,k}^{(0)} \end{bmatrix},$$

and

$$\mathbf{Q}_{\hat{\alpha}\hat{\alpha},m,k}^{(\tau)} = \frac{1}{2} \begin{bmatrix} \text{diag}(\Re(\rho_{m,k}^{(\tau)})) & -\text{diag}(\Im(\rho_{m,k}^{(\tau)})) \\ \text{diag}(\Im(\rho_{m,k}^{(\tau)})) & \text{diag}(\Re(\rho_{m,k}^{(\tau)})) \end{bmatrix},$$

$$\rho_{m,k}^{(\tau)} = \hat{\beta}_{m,k} \left[\hat{\gamma}_{m,k}^{(l)} \rho_{m,k}^{(l,\tau)} \right]_{l=1}^L,$$

such that $\rho_{m,k}^{(l,\tau)} \in [0, 1]$ is the time correlation coefficient for a limited period of time. In our simulations we consider ideal $\hat{\beta}_{m,k}$, that is, $\hat{\beta}_{m,k} = \beta_{m,k}$, and we estimate the power fraction as $\hat{\gamma}_{m,k}^{(l)} = 1/L$ for $K_R = 0$, $\hat{\gamma}_{m,k}^{(1)} = \frac{K_R}{K_R+1}$ (LoS) and $\hat{\gamma}_{m,k}^{(l \neq 1)} = \frac{1}{K_R+1} \frac{1}{L-1}$ (NLoS) for $K_R > 0$. Regarding the state covariance $\mathbf{Q}_{\mathbf{u}_{\bar{\mathbf{x}}}\mathbf{u}_{\bar{\mathbf{x}}},m,k}$, processing the real and imaginary parts of $\hat{g}$ separately, we assume that $\mathbf{Q}_{\mathbf{u}_{\bar{\mathbf{x}}}\mathbf{u}_{\bar{\mathbf{x}}},m,k} =$

updating the estimates with reduced computational cost and training sequence length, while still relying on initial estimates from methods such as OMP or MUSIC.

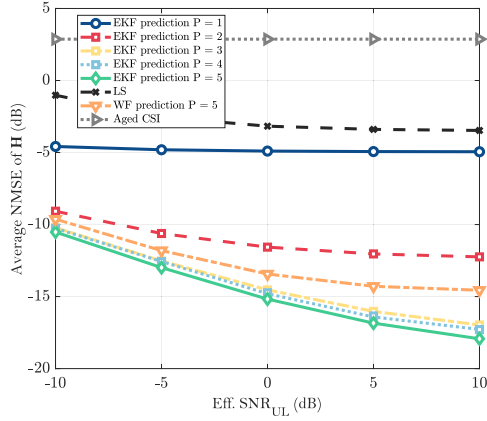


Fig. 6. NMSE of the channel matrix for Jake's isotropic scattering model and different P -orders as a function of the uplink training effective SNR.

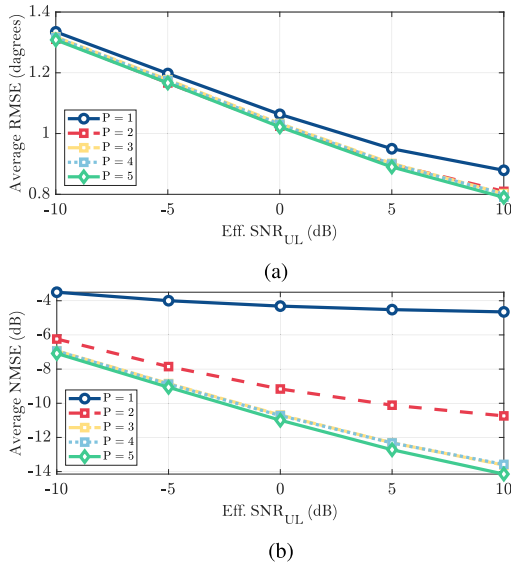


Fig. 7. MSE of EKF predictor for Jake's isotropic scattering model and different P -orders as a function of the uplink training effective SNR. (a) Path angles error. (b) Path gains error.

$\text{diag}([\sigma_{\hat{\phi}_A}, \sigma_{\hat{\phi}_D}, \sigma_{\hat{\theta}_A}, \sigma_{\hat{\theta}_D}, \sigma_{\hat{g}_{m,k}/2}, \sigma_{\hat{g}_{m,k}/2}])$, such that σ is the vector of the L variances of a single parameter and $\sigma_{\hat{g}_{m,k}}^{(l)2} = \hat{\beta}_{m,k} \hat{\gamma}_{m,k}^{(l)} \sigma_{\alpha,m,k}^{(l)2}$, where $\sigma_{\alpha,m,k}^{(l)2}$ is process variance (13) from the method in [29], and that the estimated angular variances are considered ideal. Finally, we configure the noise correlation of each AP as $\mathbf{Q}_{\mathbf{u}_{\bar{y}}\mathbf{u}_{\bar{y}},m} = \frac{1}{K} \sum_k \beta_{m,k} \mathbf{I}_{2D_{\bar{y}}}$ for simulations with controlled SNR (Figs. 6, 7, and 9) and $\mathbf{Q}_{\mathbf{u}_{\bar{y}}\mathbf{u}_{\bar{y}},m} = \frac{\sigma_{\text{UL}}^2}{2\tau_p \theta_p} \mathbf{I}_{2D_{\bar{y}}}$ for simulations with fixed receiver noise power. Let SNR_{UL} be the effective SNR on a linear scale, that considers the ideal array gain and the received power. For high SNR values, a small bias can be added to the noise power $\mathbf{Q}_{\mathbf{u}_{\bar{y}}\mathbf{u}_{\bar{y}},m}$ to avoid ill conditioning, for both EKF and WF.

IV. NUMERICAL RESULTS

In this section, we numerically evaluate the performance of the EKF predictor in estimating the general channel and the impact on downlink transmission rate in CF networks under

LoS and moderate NLoS conditions. We analyzed three parameters that influence the channel coherence time, as shown in Fig. 4: normalized maximum Doppler frequency shift (f_{ND}) of isotropic fading, angular dispersion control parameter (κ) of non-isotropic fading and the strength of the LoS component of Rician fading (K_R). Since the Doppler shift in mmWave bands is 10 times greater than in sub-6GHz bands, the performance loss due to channel aging may be greater for networks operating in this frequency range. Therefore, we focused our study on the mmWave bands. The influence of the training SNR on the prediction of the CSI and the pilot spacing on the downlink rate will also be analyzed. We use normalized mean square error (NMSE) as prediction performance indicator for the $\mathbf{H}$ -channel matrix and for the channel gains, and root mean square error for the angles. The errors in the first five slots, which are related to the filter convergence phase, were disregarded.

A. Large-Scale Fading Model

In CF networks, due to the low installation heights of the APs, the terminals can be closer to the users compared to cell-based networks, so that common obstacles create correlated shadowing coefficients and scatterer clusters. Such a correlation can have a considerable impact on SE in CF networks [3], [9]. We adopted the spatial consistency model as in [31], which creates a continuous shadowing map for the simulation environment based on stationary and isotropic Gaussian random field with zero-mean and exponentially decaying spatial correlation, such that for each point of the location grid there is a defined shadow fading coefficient. Based on [31] and [49], we retrieve from the shadowing map the shadowing coefficients $\mathcal{S}[\mathbf{r}_m]$ at the discrete location $\mathbf{r}_m$ of the m -th AP and $\mathcal{S}[\mathbf{r}_k]$ at the discrete location $\mathbf{r}_k$ of the k -th UE, such that the correlated shadowing coefficient between m -th AP and k -th UE is given by:

$$\chi_c(\mathbf{r}_m, \mathbf{r}_k) = \frac{1 - \exp\left(-\frac{d_{m,k}}{d_{\text{corr}}}\right)}{\sqrt{1 + \exp\left(-\frac{d_{m,k}}{d_{\text{corr}}}\right)}} \sqrt{0.5} (\mathcal{S}[\mathbf{r}_m] z_m + \mathcal{S}[\mathbf{r}_k] z_k), \quad (27)$$

such that

$$\mathbb{E}\{\mathcal{S}[\mathbf{r}_m] \mathcal{S}[\mathbf{r}_k]\} = \sigma_{\text{SF}}^2 \exp\left(-\frac{d_{m,k}}{d_{\text{corr}}}\right),$$

where σ_{SF}^2 is the shadowing factor and $d_{m,k} = \|\mathbf{r}_m - \mathbf{r}_k\|$ is the distance between m -th AP and k -th UE. d_{corr} is a correlation distance which depends on the size of obstructing objects.

Consider the L -scatterer channel model of an AP-UE pair formed by L fixed dominant paths. We assume a priori knowledge [32], [33] of the number of paths and their angles (AoA and AoD) as well as their large scale gains. Methods for estimating path angles are extensive and have been covered in existing papers, such as in [50], [51], [52], [53]. Inspired by [15], aiming to obtain spatial-angular similarity between nearby UEs, which results in beam conflict problems, a given number of spatial clusters were randomly generated by the environment, with a predefined minimum distance between them and between APs

and UEs. By drawing an ellipsoid of adjustable dimensions around the AP-UE pair, the L paths are randomly chosen from the scatterer clusters surrounded by the ellipsoid so that the smaller its volume, the greater the spatial correlation. Then, based on the positions of the scatterer clusters, the average AoA and AoD parameters are calculated. The distribution of the powers of the paths follows as in [54], such that the total power share for a path is given by the factor:

$$\gamma^{(l)} = \frac{\gamma'_l}{\sum_{l=1}^L \gamma'_l}, \quad (28)$$

such that

$$\gamma'_l = U_l^{r_\tau - 1} 10^{-0.1z_l}, \quad U_l \sim \mathcal{U}[0, 1], \quad z_l \sim \mathcal{N}(0, \zeta^2),$$

where r_τ and ζ are heuristic constant parameters [54].

The log-distance omnidirectional path loss model for a reference distance of 1 m is given by:

$$\text{PL}_{m,k} = \chi_a + \chi_b 10 \log_{10}(d_{m,k}) + \chi_c(\mathbf{r}_m, \mathbf{r}_k) \quad [\text{dB}], \quad (29)$$

where χ_c is the shadowing given by (27) and parameters χ_a (path loss at a reference distance) and χ_b (path loss exponent) are frequency and environment dependent whose reference values for mmWave are found in [54]. Then, the large-scale fading between the m -th AP and the k -th UE is defined as:

$$\beta_{m,k} = 10^{-\frac{\text{PL}_{m,k}}{10}}. \quad (30)$$

B. Time Autocorrelation Function

In the most general case in relation to the models frequently used in the related literature, the time ACF is spatially dependent, so that f_{ND} , distance $d_{m,k}$, the scatterer cluster angles $\phi_{\text{A(D)},m,k}^{(l)}$ and the UE velocity direction $\angle \mathbf{v}_k$ are parameters of the function.³ For simplicity, we consider that the angular elevation components do not influence the time ACF. We made the following considerations for formulating our non-isotropic model: stationary AP surrounded by scatterers in a radius r_{AP} , mobile UE surrounded by scatterers, single antenna⁴ and single-bounced (SB) path. Based on the 2D GBSM developed in [41], we define time ACF non-isotropic of the uplink channel small-scale gain α as:

$$Q_{\alpha\alpha,m,k}^{(\tau)} = \sum_{l=1}^{L'} \gamma_{m,k}^{(l)} \rho_{\text{SBAP},m,k}^{(l,\tau)} + \sum_{l=L'+1}^L \gamma_{m,k}^{(l)} \rho_{\text{SBUE},m,k}^{(l,\tau)}, \quad (31)$$

where $\rho_{\text{SBAP},m,k}^{(l,\tau)}$ and $\rho_{\text{SBUE},m,k}^{(l,\tau)}$ are given by (32) and (33) shown at the bottom of this page, corresponding to the SB path of

the AP-side scatterer cluster directly to the UE, and the SB path of the AP directly to the UE-side scatterer cluster, respectively. In those equations, $I_0(\cdot)$ is the zero-order modified Bessel function of the first kind and $\kappa_{\text{AP(UE)}}$ is the control parameter for the spread of the scatterers within each scatterer cluster coming from the von Mises distribution. Non-isotropic small-scale gains are generated with AR(200) and bias value of $\epsilon = 10^{-4}$ [29] in our simulations.

Jake's isotropic scattering model is a particular case of (31) when scatterers are not considered around the AP and when $\kappa_{\text{UE}} = 0$, such that scatterers are considered to uniformly surround the UE so that the angular spread is 2π (assumed spatially uncorrelated Rayleigh or Rician fading). In this case, $\rho_{\text{SBUE},m,k}^{(l,\tau)} = \rho_k^{(l,\tau)} = J_0(2\pi f_{\text{ND},k}\tau)$ [43], where $J_0(\cdot)$ is the zero-order Bessel function of the first kind. Note that the classic isotropic model depends, physically, only on the maximum velocity of the environment, thus being a model that simplifies the analysis. Aiming at a more precise fit, isotropic small-scale gains are generated with the IDFT method [42].

C. Parameters and Setup

Given a certain CF network, we consider a subset of $M = 20$ APs and $K = 8$ UEs operating in mmWave band under moderate NLoS and isotropic fading (Jake's model), unless otherwise stated. Each AP is equipped with a 4×4 UPA and $N_{\text{RF}}^{\text{AP}} = 8$ RF chains, transmitting $N_s = 8$ streams based on DFT codebook. UEs are equipped with a 4×4 UPA and $N_{\text{RF}}^{\text{UE}} = 4$ RF chains, receiving in single-stream mode. To keep training within 4 pilot symbols, we use half of the UE beam codebook, such that beams are chosen uniformly. Therefore, we make $B_F = 16$ and $B_W = 8$. In our simulated controlled environment, the moderate NLoS state, related to partial obstruction, is defined for $\chi_a = 61.4$ dB, $\chi_b = 2.68$ dB, $\sigma_{\text{SF}} = 7.0$ dB and $d_{\text{corr}} = 50.0$ m (correlated shadowing), with a minimum distance between APs and UEs of 40 m and maximum of 150 m, so that the probability of LoS is less than 50%, the probability of outage is zero and the path loss varies approximately between 80.0 dB to 145.0 dB,

³In practical scenarios, channel measurement techniques, such as sounding, measurement campaigns and out-of-band information can capture the real parameters of propagation paths and terminals, such as number of resolvable paths, delay, nominal AoAs and AoDs, inertial information and statistics.

⁴A single small-scale gain for the entire antenna array according to (3), such that the cross-correlation between antennas is simplified by steering vector, separating channel time aging and spatial correlation effects. Also, we neglect the small angular variation in the temporal correlation.

$$\rho_{\text{SBAP},m,k}^{(l,\tau)} = \exp(-j2\pi f_{\text{ND},k} \cos(\angle \mathbf{v}_k) \tau) \frac{I_0 \left(\sqrt{\kappa_{\text{AP}}^2 + j4\pi \kappa_{\text{AP}} \frac{r_{\text{AP}}}{d_{m,k}} f_{\text{ND},k} \sin(\phi_{\text{D},m,k}^{(l)}) \sin(\angle \mathbf{v}_k) \tau} - \left(2\pi \frac{r_{\text{AP}}}{d_{m,k}} f_{\text{ND},k} \sin(\angle \mathbf{v}_k) \tau \right)^2 \right)}{I_0(\kappa_{\text{AP}})}. \quad (32)$$

$$\rho_{\text{SBUE},m,k}^{(l,\tau)} = \frac{I_0 \left(\sqrt{\kappa_{\text{UE}}^2 + j4\pi \kappa_{\text{UE}} f_{\text{ND},k} \cos(\phi_{\text{A},m,k}^{(l)} - \angle \mathbf{v}_k) \tau} - (2\pi f_{\text{ND},k} \tau)^2 \right)}{I_0(\kappa_{\text{UE}})}. \quad (33)$$

TABLE III
OTHER DEFAULT SIMULATION PARAMETERS

Type	Parameter
Numerology [55]	$f_c = 28.0$ GHz (FR2), Bandwidth = 200.0 MHz, $T_s = 8.335 + 0.59 = 8.925$ μ s, $\tau_s = 14$ symb., $\tau_{\text{slot}} = 80$ time-slots.
Channel	$L = 4$, $\zeta = 4.0$, $r_\tau = 2.8$, $\tau_\delta = 14$ symb., $\ \mathbf{v}_k\ = 40.0$ km/h, $\angle \mathbf{v}_k$ random, $\sigma_{\phi, m, k}^{(l)} = 0.1^\circ$, $\sigma_{\theta, m, k}^{(l)} = 0.01^\circ$.
Environment	AP array height = 15.0 m, UE array height = 1.5 m, Coverage area = 250.0×250.0 m ² , Number of clusters = 250, Minimum distance between APs = 10.0 m, Minimum distance between clusters = 5.0 m, Number of simulations = 25.
Noise	Noise power spectral density $N_0 = -173.8$ dBm, Noise figure = 7.0 dB, Downlink data effective power $\rho_d = 17.1$ dB, Uplink training effective power $\rho_p = 14.1$ dB.

where we are inspired by the values of the measurement campaign in [54]. The other simulation parameters are summarized in Table III.

D. Results and Discussions

1) *Prediction Error Analysis*: at first, EKF adjusts the channel gains with a blind prediction model, built with the AR(P) process from (17), based on a received pilot sequence and its estimation model, given by (20). The time ACF adjustment depends on P -order. It is important to note that we correspond each EKF tap to a time-slot, so that the filter convergence time with P -order would last at least P time-slots. Furthermore, the dimensions of the state variable, and consequently the computational complexity of the EKF and WF, depend on the value of P . Therefore, very high orders can be disadvantageous in practice. Thus, it is important to evaluate the influence of the P -order on the NMSE. Based on the values in Table III, we set the normalized Doppler shift of the EKF at $f'_{\text{ND}} = 0.13$.

In Fig. 5 we evaluated the prediction error and transmission rate as a function of the P order for EKF and linear WF (details the implementation of WF are shown in Appendix A). For an isotropic NLoS channel model, we observed that increasing the order $P > 5$ has little influence on the error and rate, results showing between $P = 5$ and $P = 10$ a difference of 6.0 Mbps of rate, approximately. Aiming at low computational complexity, we use maximum order $P = 5$.

2) *SNR Analysis*: Considering the described moderate NLoS conditions and pilot power of -10.0 dB, we analyzed the impact of training effective SNR per AP-UE pair on the performance of the EKF predictor in the range of -10.0 dB to 10.0 dB. Fig. 6 shows the NMSE of channel matrix prediction performed at n_i for the instant $n_i + \tau_\delta$, $P \in \{1, 2, 3, 4, 5\}$ and $\epsilon = 10^{-3}$. We compare with the benchmarks non-predictive method based on LS [38] and linear WF predictor, both with the same overhead, and we also plot the error of the ideal channel at simulation instant zero, represented by “Aged CSI”, which expresses how

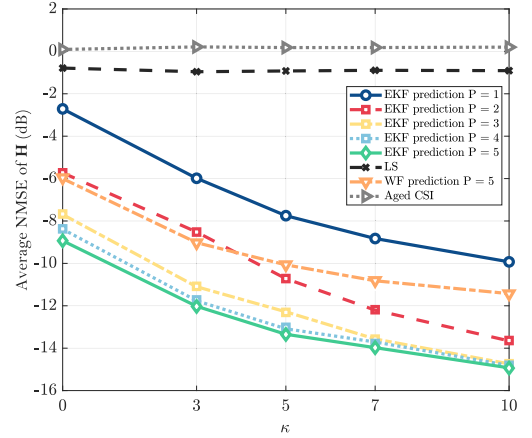


Fig. 8. Channel matrix error of EKF predictor for different P -orders as a function of the κ parameter.

much the channel ages within a frame. We note that under channel simulation conditions, the LS method presents a poor performance, sometimes worse than CSI without update, as it is sensitive to the three types of errors present related to: low SNR, channel aging and insufficient measurements. The WF, often referred to as a linear minimum mean square error predictor, performs slightly better than LS, but it may present inferior performance to the EKF of order $P = 5$. The following causes can influence the performance loss of the WF predictor: estimation error of the channel correlation matrix, angular variation of the paths and insufficient measurements. In addition, the computational cost of calculating the inverse correlation matrix of the training (40) is greater than the cost of inversely calculating the EKF gain (24), since the WF propagating variable is the training sequence. In turn, the EKF predictor can bring significant improvements to channel estimation using AR models with order higher than the conventional $P = 1$, in addition to being able to estimate propagation parameters (Fig. 7), in comparison with LS and WF, that are useful in modern applications. As for P -order, by the gap between the curves, we note that the errors do not decrease linearly with P , so that the SNR influences the difference in performance between the P -orders of the AR model. Our simulations show that $P = 5$ is enough to handle time ACF based on isotropic model to $f'_{\text{ND}} = 0.13$ and $P = 3$ can be adopted aiming at a better compromise between performance and computational burden.

In Fig. 8 we analyze error in relation to the angular dispersion control parameter κ , considering that $\kappa = \kappa_{\text{UE}} = \kappa_{\text{AP}}$. We adopt random UE travel directions and half of the paths of SB-AP type and the other of SB-UE type. We set $\tau_\delta = 28$ and the training noise power according to the values in Table III. We note that the NMSE difference between orders tends to decrease with increasing κ ,⁶ so that $P = 3$ can be accurately used for directive channels. This behavior is expected by the insights taken from the absolute time correlation curves at Fig. 4(b),

⁵Values are converted to standard units: second, meter per second, radian, hertz and watt.

⁶Isotropic channel for $\kappa = 0$ in (32) considers scatterers around the AP, which is different from Jake's model.

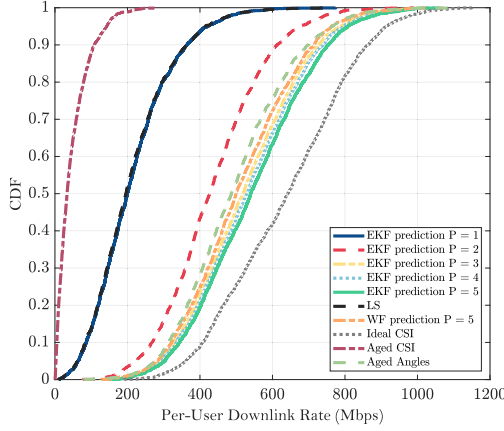


Fig. 9. CDF per-user downlink rate with bandwidth of 200 MHz, Jake's model and 10 dB uplink training effective SNR.

in which the more directive the propagation conditions, the greater the channel coherence time. Also, we notice that the performance limit of the WF predictor for long coherence time related to high κ values can be smaller than the EKF predictor. Due to the diverse configuration of non-isotropic models, which depend on many joint parameters such as scatterer cluster angles, velocity module and phase, $d_{m,k}$, r_{AP} , κ_{AP} and κ_{UE} , aiming at practical applications, it is more robust to consider $P > 2$ order AR models ($P = 3$ for good compromise between performance and computational burden) in this case than $P = 2$.

3) *Rate Analysis:* In Fig. 9 we compared the performance in terms of downlink rate of ZF precoders determined with CSI predicted by EKF and compared with the LS, WF and with ideal CSI (without estimation errors and aging), all with the same overhead. We also plot the curve with ZF precoders calculated with ideal CSI but without updating the angular parameters, labeled as “Aged Angles”. We note that the performance improvement observed in the NMSE curves is also reflected in the rate, where the ZF precoder with $P > 2$ shows significant improvements compared to $P \leq 2$ and that $P = 3$ is enough to mitigate the effect of channel aging to $f'_{ND} = 0.13$. We also note that the small angular perturbations of NLoS paths in time can contribute to the effect of channel aging on rate deterioration if neglected. This is due to the beam alignment error in (10), and consequently the loss of beamforming gain, arising from the outdated angles. Such impact is greatest for narrow beams in large antenna arrays [33] and beamsteering codebooks [56].

Fig. 10 presents the impact of the parameter κ of the non-isotropic model on the downlink rate. We observed behavior similar to that observed in the error curves, where the performance difference decreases in relation to the P -order as the directivity increases, such that $P \geq 3$ orders have little impact on the rate for highly non-isotropy conditions.

4) *Normalized Doppler Shift Analysis:* In Figs. 11 and 12 we analyze the performance of the system for different values of normalized Doppler shift. The normalized effective value f'_{ND} is proportional to: f_c , $\|\mathbf{v}_k\|$, T_s and τ_δ . Considering f_c and $\|\mathbf{v}_k\|$ fixed, f'_{ND} can be reduced by decreasing the values of T_s and

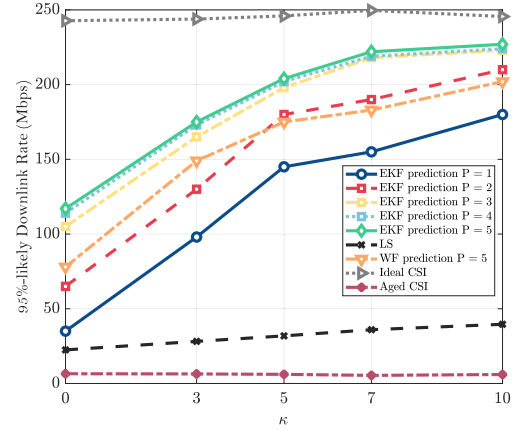


Fig. 10. Influence of the angular dispersion on the 95%-likely per-user downlink rate.

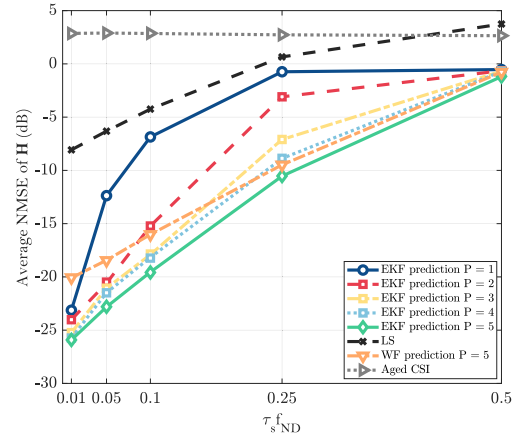


Fig. 11. NMSE of the channel matrix for different P -orders in relation to the normalized Doppler shift f'_{ND} .

τ_δ . However, the reduction of T_s is limited by bandwidth. As mentioned before, we assume in this paper that $\tau_\delta = \tau_s$, so the longer the transmission block length is, the higher f'_{ND} will be.

Fig. 11 shows the NMSE of the EKF predictor for $\tau_\delta = \tau_s = 14$ symbols, different values of f_{ND} and Jake's isotropic model. We note three important points: the extreme values and the central value. For the smallest value of f_{ND} the filter order does not bring improvements. This is because for low values of f_{ND} the time-correlation drops very little within the delay range (Fig. 4(a)), so that $P = 1$ is sufficient. Also note that the order P does not influence the highest value of f_{ND} . In this condition, the correlation between two consecutive channel samples varies greatly, thus losing the smoothness property. For central values, note that the P order improves performance, and can be set from 2 to 5. In Fig. 12 we analyze the downlink rate for $f_{ND} = 0.0093$, according to the data in Table III, with different transmission block length τ_s , such that $f'_{ND} = \tau_s f_{ND}$. There is a trade-off in rate between channel EKF predictor accuracy and training overhead. Note that the highest rate of 300 Mbps is achieved for $\tau_s = 12$, however, the 3GPP NR default value $\tau_s = 14$ and $P > 2$ can satisfy the trade-off for the

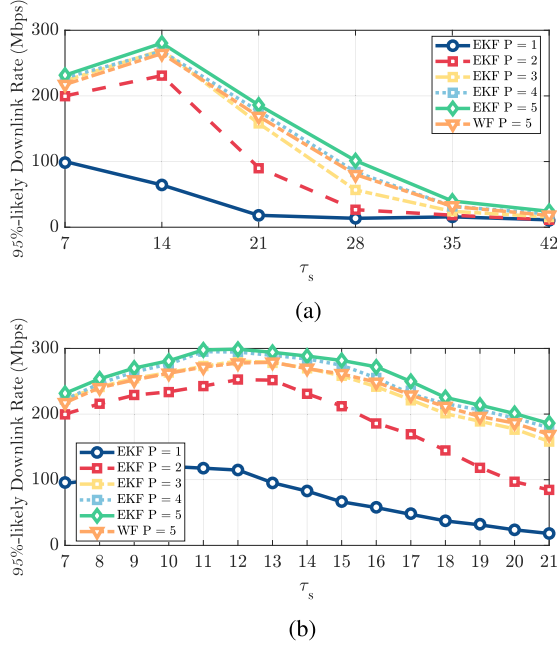


Fig. 12. 95%-likely of downlink rate in relation to the transmission block length for different P -orders. (a) Analysis for every 7 block symbols. (b) Detailed analysis for every 1 symbol.

setup adopted. Therefore, setting $P > 2$ guarantees good performance over a wide Doppler shift range, so that the choice $P = 3$ has the best compromise between accuracy and computational complexity.

5) *Rician K-Factor Analysis*: So far, we have analyzed the performance of the proposed filter in the NLoS multipath⁷ environment, considered the worst case. Let us now consider the influence of the LoS component with the following deterministic time ACF in the zero initial phase:

$$\rho_{\text{LOS},k}^{(\tau)} = \exp(j2\pi f_{\text{ND},k}\tau \cos(\angle \mathbf{v}_k)), \quad (34)$$

so the uplink ACF of the LoS multipath fading channel is:

$$Q_{\alpha\alpha,m,k}^{(\tau)} = \frac{1}{1 + K_R} \left(K_R \rho_{\text{LOS},k}^{(\tau)} + \sum_{l=2}^L \gamma_{m,k}^{(l)} \rho_{\text{NLOS},m,k}^{(l,\tau)} \right), \quad (35)$$

where $\rho_{\text{NLOS},m,k}^{(l,\tau)}$ is the time ACF of the NLoS components and K_R is the Rician K-factor which measures the power of the LoS component in relation to the total power of the NLoS components. In principle, the higher the K-factor value, the more predominant is the LoS correlation over the overall correlation. In practice, since the LoS component varies more slowly than the NLoS components, its predominance causes the absolute autocorrelation curve to slowly decrease over time, as shown in Fig. 4(c).

For analysis purposes, we assume an omnidirectional K-factor uncorrelated with the other channel parameters. In the LoS scenario, we assume: very small angular variation of the direct path, minimum horizontal distance between AP and UE

⁷Herein, multipath refers to the dominant spatially resolvable paths L , as well as the subpaths (micro-scatterers) that can implicitly compose each of them.

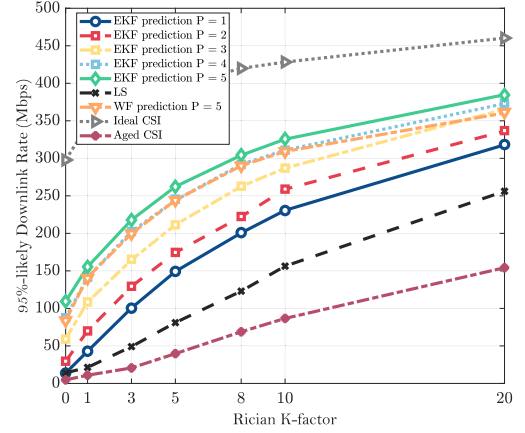


Fig. 13. Influence of the LoS component on the achievable rate for different P -order values.

of 20.0 m and maximum of 110.0 m, and path loss parameters listed in [54], with loss values varying between 60.0 dB and 135.0 dB in the simulations. Regarding the small-scale gains, we consider an isotropic scattering model for generating the gains of NLoS paths ($l = \{2, 3, \dots, L\}$), which we implement with IDFT, and the deterministic model of (34) for the LoS path ($l = 1$). We set $f_{\text{ND},k} = 0.0093$ and $\tau_s = 28$. The coefficients of the EKF transition matrix corresponding to the LoS path were obtained for $\epsilon = 10^{-2}$ in order not to generate very low values of $\sigma_{\alpha,m,k}^{(1)2}$. The results in terms of achievable rate of the precoders determined with the predicted channel are shown in Fig. 13 for the default parameters of Table III. We observe a behavior similar to that observed in κ , where the error difference between orders tends to decrease with the increase of the K-factor, so that orders $P \geq 3$ can be used with precision for strong LoS. The same behavior is also observed in the rate curves, where $P \geq 3$ has rates similar to higher orders while $P < 3$ may have lower rates. Thus, $P \geq 3$ is a good configuration for large K-factor ranges, while $P < 3$ can be adopted for strong LoS cases in order to reduce complexity.

V. CONCLUSION

In this paper, a channel parameters predictor based on EKF and P -order AR process was proposed and evaluated in order to mitigate the degradation effects of channel aging in CF networks. From our simulations, we observed that the CF MIMO system based on HBF architecture operating in downlink with Doppler normalized shift of approximately $f_{\text{ND}} \sim 0.01$, resulting from a Doppler frequency of 1.0 kHz and a prefix plus symbol period of 8.9 ns, cannot maintain communication with CSI update period of one time-frame. Furthermore, we observe that the CSI estimation must have a predictive nature, in order to predict it for the next transmission block based on the current training, since the inherent delay size of the network is sufficient for estimates of short coherence time channels to age. Under these conditions, the EKF deals very well with the channel aging, HBF architectures, spatially uncorrelated Rayleigh and Rician fading, enabling the prediction of the channel with space pilot reduction, being able

to operate over a wide range of Doppler frequencies. From our research with isotropic and non-isotropic models, we conclude that models of order $P \geq 3$ can increase the robustness of channel prediction methods against the diversity of propagation conditions in CF networks. Also, we observed that orders $P \leq 5$ of the AR process are sufficient to mitigate the effects of channel aging in Rayleigh and Rician fading environment. As general conclusions, channel conditions, such as path clustering, SNR, Doppler shift, K-factor and κ , influence the P -order, so that $P > 2$ can provide significant improvements in mitigating the impact of channel aging. Furthermore, we conclude from our methodology that $P = 3$ has a good compromise between performance and computational burden that holds under different channel conditions. The proposed model can also be adapted to mitigate the effect of the phase noise and pilot contamination, which is a task for future work.

APPENDIX A WIENER PREDICTOR

We extend the method in [11] to match our configuration. The optimal P -th order Wiener linear predictor of the $\mathbf{H}_{\text{UL},m,k}^{(n_i)}$ for a set of input temporal measurements $\bar{\mathbf{y}}_{m,k}^{(n_i)} := [\bar{\mathbf{y}}_{m,k}^{(n_i)\dagger}, \bar{\mathbf{y}}_{m,k}^{(n_i-\tau_\delta)\dagger}, \dots, \bar{\mathbf{y}}_{m,k}^{(n_i-\tau_\delta P)\dagger}]^\dagger \in \mathbb{C}^{D_{\bar{\mathbf{y}}}(P+1) \times 1}$ is given by:

$$\hat{\mathbf{h}}_{\text{UL},m,k}^{(n_i+\tau_\delta)} = \bar{\mathbf{V}}_{m,k} \bar{\mathbf{y}}_{m,k}^{(n_i)} \in \mathbb{C}^{N_{\text{AN}}^{\text{AP}} N_{\text{AN}}^{\text{UE}} \times 1}, \quad (36)$$

such that

$$\bar{\mathbf{V}}_{m,k} = \mathbf{Q}_{\bar{\mathbf{h}},m,k}^{(1)} \mathbf{Q}_{\bar{\mathbf{y}},m,k}^{(0)-1} \in \mathbb{C}^{N_{\text{AN}}^{\text{AP}} N_{\text{AN}}^{\text{UE}} \times D_{\bar{\mathbf{y}}}(P+1)}, \quad (37)$$

where cross-correlation between the desired channel and the training signals is

$$\mathbf{Q}_{\bar{\mathbf{h}},m,k}^{(1)} = \left[\mathbf{Q}_{\bar{\mathbf{h}},m,k}^{(\tau_\delta)}, \mathbf{Q}_{\bar{\mathbf{h}},m,k}^{(2\tau_\delta)}, \dots, \mathbf{Q}_{\bar{\mathbf{h}},m,k}^{(\tau_\delta P + \tau_\delta)} \right] \otimes \mathbf{L}^\dagger, \quad (38)$$

where $\mathbf{Q}_{\bar{\mathbf{h}},m,k}$ is the covariance matrix of the uplink channel, estimated here by the average:

$$\mathbf{Q}_{\bar{\mathbf{h}},m,k}^{(\tau)} = \hat{\beta}_{m,k} \sum_{l=1}^L \hat{\gamma}_{m,k}^{(l)} \rho_{m,k}^{(l)} [\tau] \mathbf{h}_{\text{UL},m,k}^{(l,0)} \mathbf{h}_{\text{UL},m,k}^{(l,0)\dagger}, \quad (39)$$

such that

$$\mathbf{h}_{\text{UL},m,k}^{(l,0)} = \text{vec} \left(\mathbf{a}_{\text{AP}} \left(\hat{\theta}_{\text{A},m,k}^{(l,0)}, \hat{\phi}_{\text{A},m,k}^{(l,0)} \right) \mathbf{a}_{\text{UE}}^\dagger \left(\hat{\theta}_{\text{D},m,k}^{(l,0)}, \hat{\phi}_{\text{D},m,k}^{(l,0)} \right) \right).$$

We consider the a priori knowledge of $\beta_{m,k}$, the initial angles of paths, number of dominant paths and time ACF, as considered in the EKF. The autocorrelation matrix of training signals $\mathbf{Q}_{\bar{\mathbf{y}},m,k}^{(0)} \in \mathbb{C}^{D_{\bar{\mathbf{y}}}(P+1) \times D_{\bar{\mathbf{y}}}(P+1)}$ is:

$$\mathbf{Q}_{\bar{\mathbf{y}},m,k}^{(0)} = \left(\mathbf{L} \otimes (\mathbf{P}_{m,k} \otimes \mathbf{L}^\dagger) + \mathbf{Q}_{\bar{\mathbf{u}},m,k} \right)^{-1}, \quad (40)$$

where

$$\mathbf{P}_{m,k} = [\mathbf{P}_{hj}]_{1 \leq h,j \leq P+1} \text{ and } \mathbf{P}_{hj} = \begin{cases} \mathbf{Q}_{\bar{\mathbf{h}},m,k}^{(\tau_\delta|h-j|)}, & \text{if } h \geq j \\ \mathbf{Q}_{\bar{\mathbf{h}},m,k}^{(\tau_\delta|h-j|)\dagger}, & \text{if } h < j \end{cases}.$$

Note that $\mathbf{Q}_{\bar{\mathbf{u}},m,k}$ is a diagonal matrix just like in EKF, but with dimension $D_{\bar{\mathbf{y}}}(P+1)$. The operation of inverting a matrix of size $D_{\bar{\mathbf{y}}}(P+1) \times D_{\bar{\mathbf{y}}}(P+1)$ in (40) is responsible for the computational complexity of WF. In the above equations, interference due to pilot reuse was not considered.

REFERENCES

- [1] W. Jiang, B. Han, M. A. Habibi, and H. D. Schotten, "The road towards 6G: A comprehensive survey," *IEEE Open J. Commun. Soc.*, vol. 2, pp. 334–366, 2021.
- [2] S. Elhoushy, M. Ibrahim, and W. Hamouda, "Cell-free massive MIMO: A survey," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 1, pp. 492–523, Firstquarter 2022.
- [3] H. Q. Ngo, A. Ashikhmin, H. Yang, E. G. Larsson, and T. L. Marzetta, "Cell-free massive MIMO versus small cells," *IEEE Trans. Wireless Commun.*, vol. 16, no. 3, pp. 1834–1850, Mar. 2017.
- [4] E. Björnson et al., "Massive MIMO networks: Spectral, energy, and hardware efficiency," *Found. Trends Signal Process.*, vol. 11, no. 3–4, pp. 154–655, 2017.
- [5] Z. Chen and E. Björnson, "Channel hardening and favorable propagation in cell-free massive MIMO with stochastic geometry," *IEEE Trans. Commun.*, vol. 66, no. 11, pp. 5205–5219, Nov. 2018.
- [6] C. Neil, M. Shafi, P. Smith, P. Dmochowski, and J. Zhang, "Impact of microwave and mmWave channel models on 5G systems performance," *IEEE Trans. Antennas Propag.*, vol. 65, no. 12, pp. 6505–6520, Dec. 2017.
- [7] A. Polegre, F. Palou, G. Femenias, and A. Armada, "Channel hardening in cell-free and user-centric massive MIMO networks with spatially correlated Ricean fading," *IEEE Access*, vol. 8, pp. 139827–139845, 2020.
- [8] A. Polegre, F. Palou, G. Femenias, and A. Armada, "New insights on channel hardening in cell-free massive MIMO networks," in *Proc. IEEE Int. Conf. Commun. Workshops*, 2020, pp. 1–7.
- [9] J. Zheng, J. Zhang, E. Björnson, and B. Ai, "Impact of channel aging on cell-free massive MIMO over spatially correlated channels," *IEEE Trans. Wireless Commun.*, vol. 20, no. 10, pp. 6451–6466, Oct. 2021.
- [10] L. Thiele, M. Olbrich, M. Kurras, and B. Matthiesen, "Channel aging effects in CoMP transmission: Gains from linear channel prediction," in *Proc. IEEE Conf. Rec. 45th Asilomar Conf. Signals, Syst. Comput.*, 2011, pp. 1924–1928.
- [11] K. T. Truong and R. W. Heath, "Effects of channel aging in massive MIMO systems," *J. Commun. Netw.*, vol. 15, no. 4, pp. 338–351, 2013.
- [12] A. K. Papazafeiropoulos, "Impact of general channel aging conditions on the downlink performance of massive MIMO," *IEEE Trans. Veh. Technol.*, vol. 66, no. 2, pp. 1428–1442, Feb. 2017.
- [13] E. Nayeibi, A. Ashikhmin, T. L. Marzetta, H. Yang, and B. D. Rao, "Precoding and power optimization in cell-free massive MIMO systems," *IEEE Trans. Wireless Commun.*, vol. 16, no. 7, pp. 4445–4459, Jul. 2017.
- [14] G. Femenias and F. Riera-Palou, "Cell-free millimeter-wave massive MIMO systems with limited fronthaul capacity," *IEEE Access*, vol. 7, pp. 44596–44612, 2019.
- [15] M. Alonzo, S. Buzzi, A. Zappone, and C. D'Elia, "Energy-efficient power control in cell-free and user-centric massive MIMO at mmWave," *IEEE Trans. Green Commun. Netw.*, vol. 3, no. 3, pp. 651–663, Sep. 2019.
- [16] L. D. Nguyen, T. Q. Duong, H. Q. Ngo, and K. Tourki, "Energy efficiency in cell-free massive MIMO with zero-forcing precoding design," *IEEE Commun. Lett.*, vol. 21, no. 8, pp. 1871–1874, Aug. 2017.
- [17] W. Jiang and H. D. Schotten, "Impact of channel aging on zero-forcing precoding in cell-free massive MIMO systems," *IEEE Commun. Lett.*, vol. 25, no. 9, pp. 3114–3118, Sep. 2021.
- [18] C. Kong, C. Zhong, A. Papazafeiropoulos, M. Matthaiou, and Z. Zhang, "Sum-rate and power scaling of massive MIMO systems with channel aging," *IEEE Trans. Commun.*, vol. 63, no. 12, pp. 4879–4893, Dec. 2015.
- [19] S. Kashyap, C. Mollén, E. Björnson, and E. G. Larsson, "Performance analysis of TDD massive MIMO with Kalman channel prediction," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, 2017, pp. 3554–3558.
- [20] A. H. El Hussein, E. P. Simon, and L. Ros, "Second-order autoregressive model-based Kalman filter for the estimation of a slow fading channel described by the Clarke model: Optimal tuning and interpretation," *Digit. Signal Process.*, vol. 90, pp. 125–141, 2019.
- [21] A. R. L. Paiva, G. Fodor, W. C. Freitas, Y. C. B. Silva, and C. F. M. e Silva, "Kalman-filter-based tracking of millimeter-wave channel parameters for V2X applications," in *Proc. IEEE Conf. Standards Commun. Netw.*, 2019, pp. 1–7.

- [22] R. Chopra and C. R. Murthy, "Data aided MSE-optimal time varying channel tracking in massive MIMO systems," *IEEE Trans. Signal Process.*, vol. 69, pp. 4219–4233, 2021.
- [23] H. Kim, S. Kim, H. Lee, C. Jang, Y. Choi, and J. Choi, "Massive MIMO channel prediction: Kalman filtering vs. machine learning," *IEEE Trans. Commun.*, vol. 69, no. 1, pp. 518–528, Jan. 2021.
- [24] R. Varshney, C. Gangal, M. Sharique, and M. S. Ansari, "Deep learning based wireless channel prediction: 5G scenario," *Procedia Comput. Sci.*, vol. 218, pp. 2626–2635, 2023.
- [25] R. Chopra, C. R. Murthy, and A. K. Papazafeiropoulos, "Uplink performance analysis of cell-free mMIMO systems under channel aging," *IEEE Commun. Lett.*, vol. 25, no. 7, pp. 2206–2210, Jul. 2021.
- [26] A. Papazafeiropoulos, "Impact of hardware impairments and channel aging mmWave heterogeneous networks," in *Proc. IEEE Wireless Commun. Netw. Conf.*, 2019, pp. 1–7.
- [27] C. D'Andrea, G. Interdonato, and S. Buzzi, "User-centric handover in mmWave cell-free massive MIMO with user mobility," in *Proc. 29th Eur. Signal Process. Conf.*, 2021, pp. 1–5.
- [28] W. Jiang and H. D. Schotten, "Deep learning-aided delay-tolerant zero-forcing precoding in cell-free massive MIMO," in *Proc. IEEE 96th Veh. Technol. Conf.*, 2022, pp. 1–5.
- [29] K. E. Baddour and N. C. Beaulieu, "Autoregressive modeling for fading channel simulation," *IEEE Trans. Wireless Commun.*, vol. 4, no. 4, pp. 1650–1662, Jul. 2005.
- [30] K. E. Baddour and N. C. Beaulieu, "Accurate simulation of multiple cross-correlated Rician fading channels," *IEEE Trans. Commun.*, vol. 52, no. 11, pp. 1980–1987, Nov. 2004.
- [31] S. Ju and T. Rappaport, "Simulating motion-incorporating spatial consistency into NYUSIM channel model," in *Proc. IEEE 88th Veh. Technol. Conf.*, 2018, pp. 1–6.
- [32] C. Zhang, D. Guo, and P. Fan, "Tracking angles of departure and arrival in a mobile millimeter wave channel," in *Proc. IEEE Int. Conf. Commun.*, 2016, pp. 1–6.
- [33] V. Va, H. Vikalo, and R. W. Heath, "Beam tracking for mobile millimeter wave communication systems," in *Proc. IEEE Glob. Conf. Signal Inf. Process.*, 2016, pp. 743–747.
- [34] Y. Huang, Q. Wu, T. Wang, G. Zhou, and R. Zhang, "3D beam tracking for cellular-connected UAV," *IEEE Wireless Commun. Lett.*, vol. 9, no. 5, pp. 736–740, May 2020.
- [35] J. Zhao, F. Gao, W. Jia, S. Zhang, S. Jin, and H. Lin, "Angle domain hybrid precoding and channel tracking for millimeter wave massive MIMO systems," *IEEE Trans. Wireless Commun.*, vol. 16, no. 10, pp. 6868–6880, Oct. 2017.
- [36] C. Wei et al., "Joint AOA-RSS fingerprint based localization for cell-free massive MIMO systems," in *Proc. IEEE 6th Int. Conf. Comput. Commun.*, 2020, pp. 590–595.
- [37] K. Wu, J. A. Zhang, and Y. J. Guo, *Joint Communication and Sensing: From Fundamentals to Advanced Techniques*. Hoboken, NJ, USA: Wiley, 2022.
- [38] J. Lee, G.-T. Gil, and Y. H. Lee, "Exploiting spatial sparsity for estimating channels of hybrid MIMO systems in millimeter wave communications," in *Proc. IEEE Glob. Commun. Conf.*, 2014, pp. 3326–3331.
- [39] W. Wu, D. Liu, Z. Li, X. Hou, and M. Liu, "Two-stage 3D codebook design and beam training for millimeter-wave massive MIMO systems," in *Proc. IEEE 85th Veh. Technol. Conf.*, 2017, pp. 1–7.
- [40] D. Tse and P. Viswanath, *Fundamentals of Wireless Communication*. Cambridge, U.K.: Cambridge Univ. Press, 2005.
- [41] R. He, B. Ai, G. L. Stüber, G. Wang, and Z. Zhong, "Geometrical-based modeling for millimeter-wave MIMO mobile-to-mobile channels," *IEEE Trans. Veh. Technol.*, vol. 67, no. 4, pp. 2848–2863, Apr. 2018.
- [42] D. J. Young and N. C. Beaulieu, "The generation of correlated Rayleigh random variates by inverse discrete Fourier transform," *IEEE Trans. Commun.*, vol. 48, no. 7, pp. 1114–1127, Jul. 2000.
- [43] J. He, T. Kim, H. Ghauch, K. Liu, and G. Wang, "Millimeter wave MIMO channel tracking systems," in *Proc. IEEE Globecom Workshops*, 2014, pp. 416–421.
- [44] L. Bai, Z. Huang, Y. Li, and X. Cheng, "A 3D cluster-based channel model for 5G and beyond Vehicle-to-Vehicle massive MIMO channels," *IEEE Trans. Veh. Technol.*, vol. 70, no. 9, pp. 8401–8414, Sep. 2021.
- [45] Y. Tsang and A. Poon, "Detecting human blockage and device movement in mmWave communication system," in *Proc. IEEE Glob. Telecommun. Conf.*, 2011, pp. 1–6.
- [46] Z. Ma et al., "Impact of UAV rotation on MIMO channel characterization for air-to-ground communication systems," *IEEE Trans. Veh. Technol.*, vol. 69, no. 11, pp. 12418–12431, Nov. 2020.
- [47] S. J. Julier and J. K. Uhlmann, "New extension of the Kalman filter to nonlinear systems," in *Proc. SPIE Conf. Signal Process., Sensor Fusion, Target Recognit.*, 1997, vol. 3068, pp. 182–193.
- [48] Z. Guo, X. Wang, and W. Heng, "Millimeter-wave channel estimation based on 2-D beamspace music method," *IEEE Trans. Wireless Commun.*, vol. 16, no. 8, pp. 5384–5394, Aug. 2017.
- [49] S. Lu, J. May, and R. J. Haines, "Efficient modeling of correlated shadow fading in dense wireless multi-hop networks," in *Proc. IEEE Wireless Commun. Netw. Conf.*, 2014, pp. 311–316.
- [50] D. Fan et al., "Angle domain channel estimation in hybrid millimeter wave massive MIMO systems," *IEEE Trans. Wireless Commun.*, vol. 17, no. 12, pp. 8165–8179, Dec. 2018.
- [51] J. Singh and S. Ramakrishna, "On the feasibility of codebook-based beamforming in millimeter wave systems with multiple antenna arrays," *IEEE Trans. Wireless Commun.*, vol. 14, no. 5, pp. 2670–2683, May 2015.
- [52] C. Zhang, D. Guo, and P. Fan, "Mobile millimeter wave channel acquisition, tracking, and abrupt change detection," 2016, *arXiv:1610.09626*.
- [53] Q. Qin, L. Gui, P. Cheng, and B. Gong, "Time-varying channel estimation for millimeter wave multiuser MIMO systems," *IEEE Trans. Veh. Technol.*, vol. 67, no. 10, pp. 9435–9448, Oct. 2018.
- [54] M. R. Akdeniz et al., "Millimeter wave channel modeling and cellular capacity evaluation," *IEEE J. Sel. Areas Commun.*, vol. 32, no. 6, pp. 1164–1179, Jun. 2014.
- [55] M. H. C. Garcia et al., "A tutorial on 5G NR V2X communications," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 3, pp. 1972–2026, 3rd Quarter 2021.
- [56] A. Alkhateeb, R. W. Heath, and G. Leus, "Achievable rates of multi-user millimeter wave systems with hybrid precoding," in *Proc. IEEE Int. Conf. Commun. Workshop*, 2015, pp. 1232–1237.